

BFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend

Complete

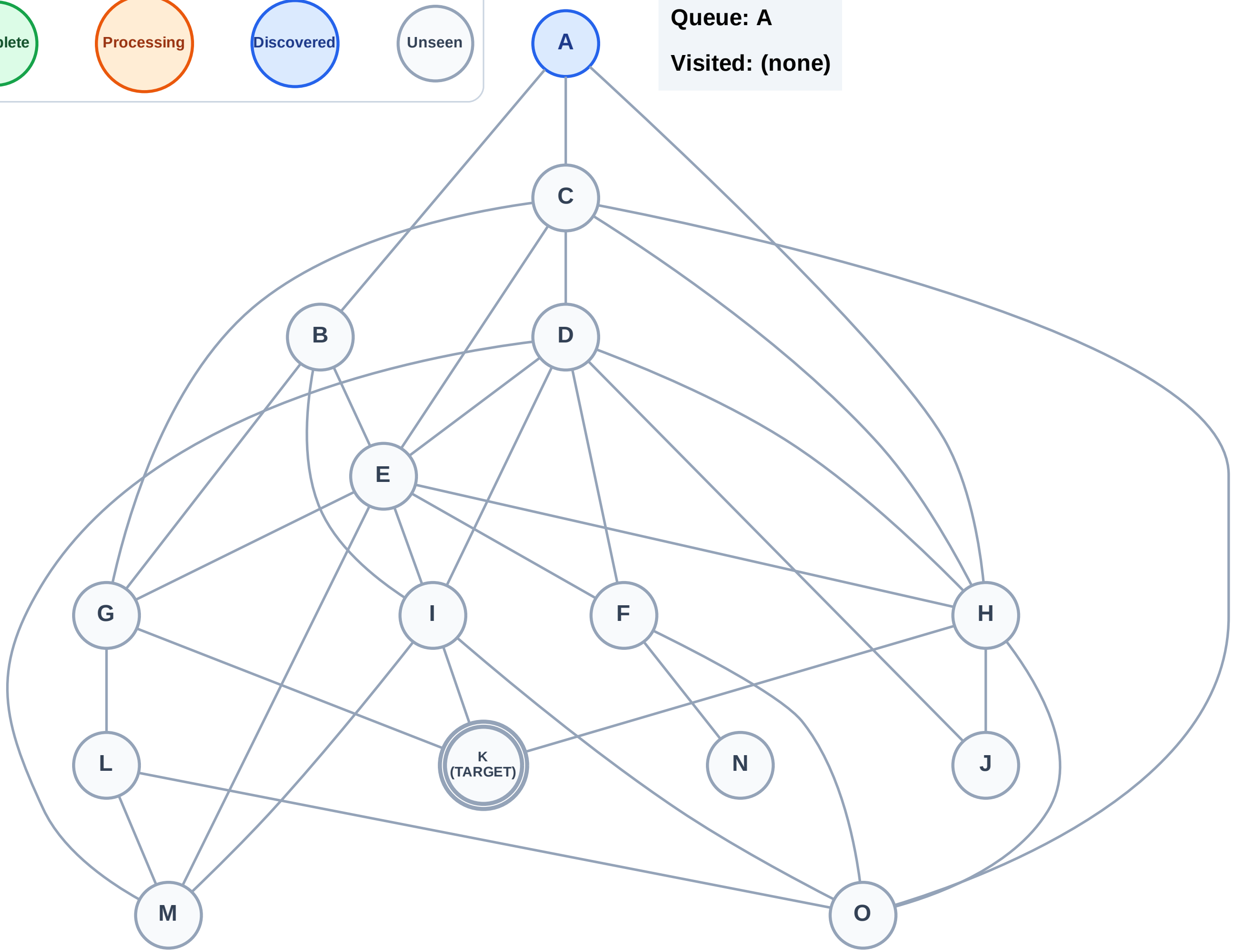
Processing

Discovered

Unseen

Queue: A

Visited: (none)



BFS Traversal

Step 2: Process node A

Legend

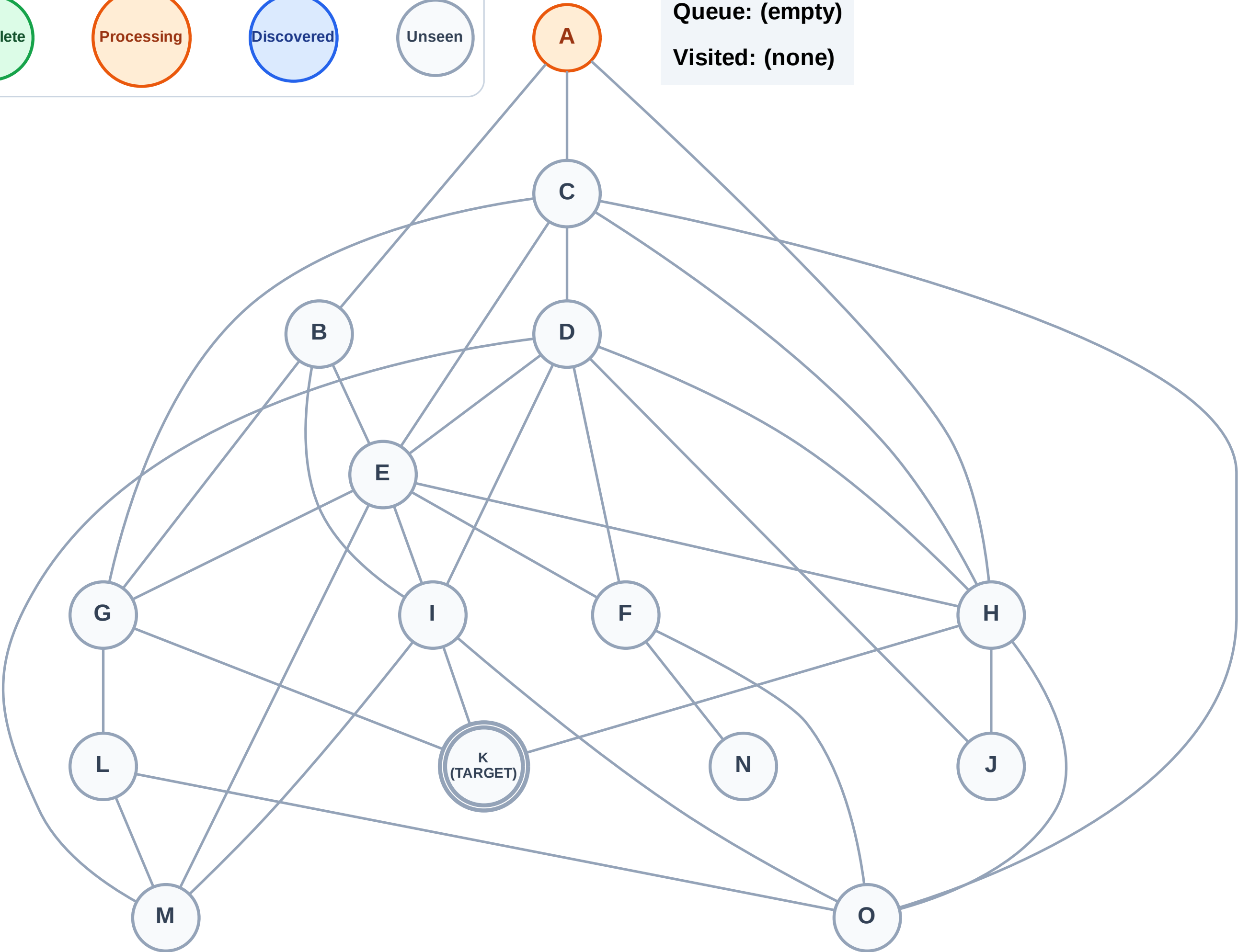
Complete

Processing

Discovered

Unseen

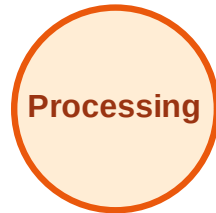
Queue: (empty)
Visited: (none)



BFS Traversal

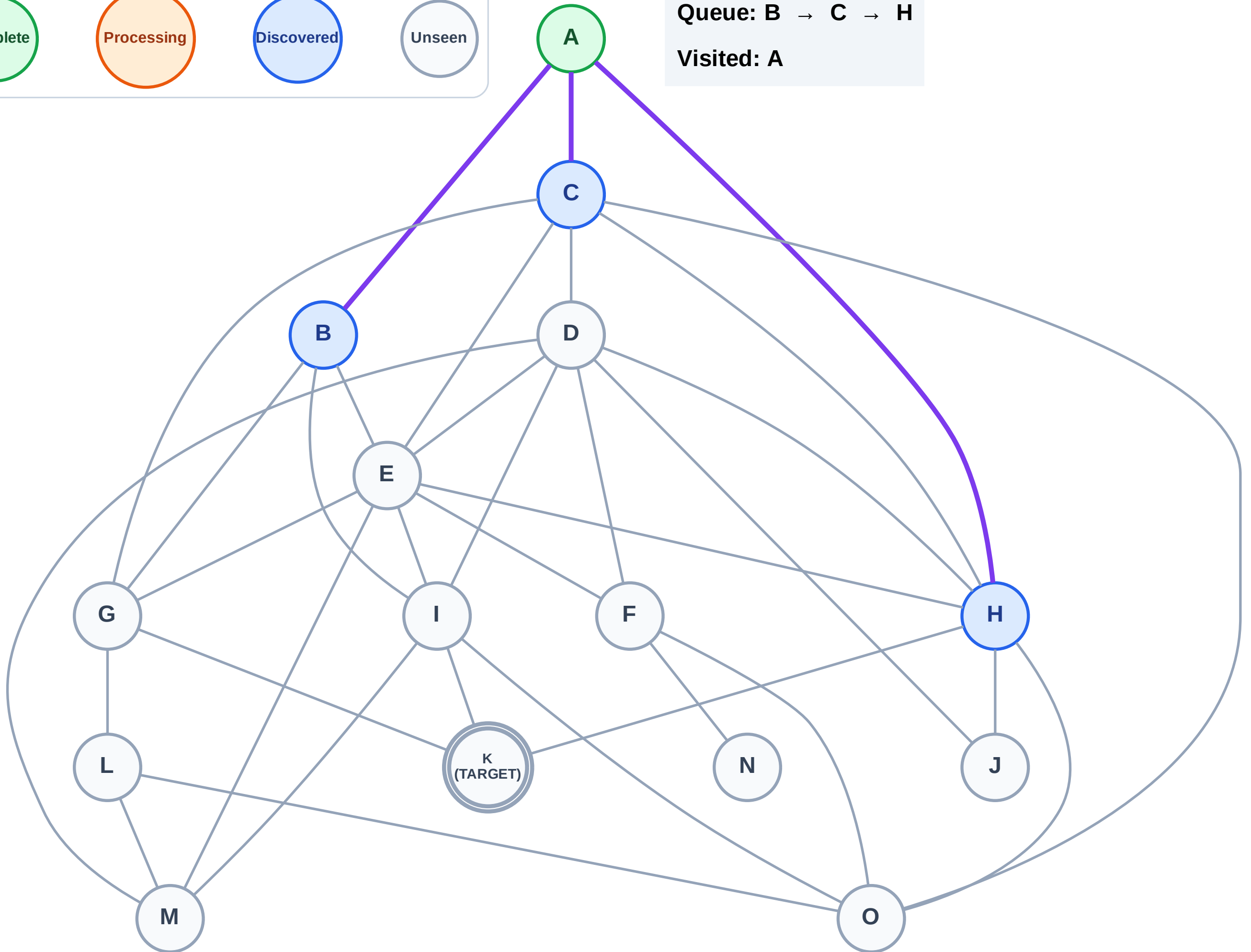
Step 3: Discover B, C, H from A

Legend



Queue: B → C → H

Visited: A



BFS Traversal

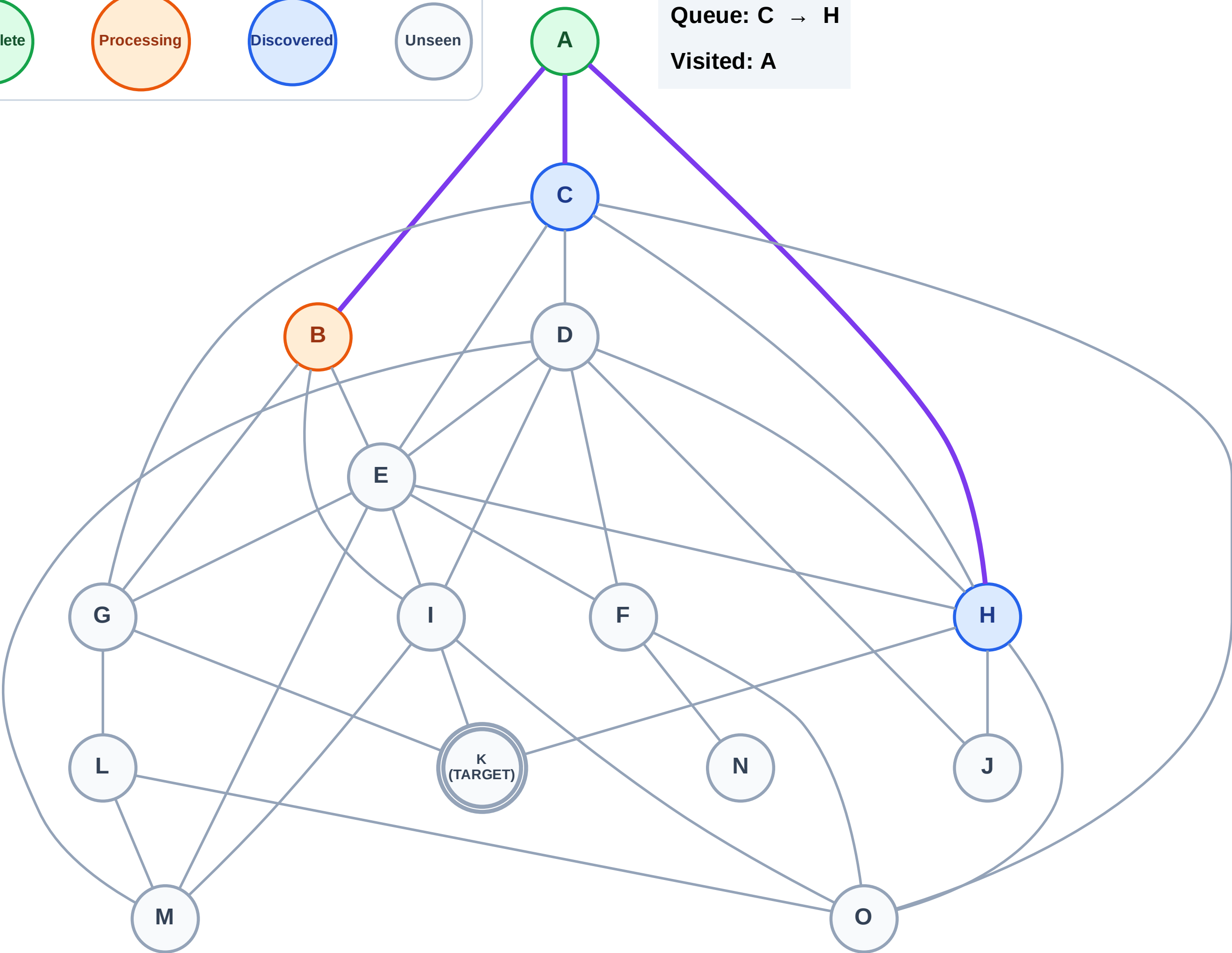
Step 4: Process node B

Legend



Queue: C → H

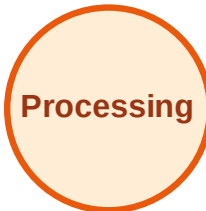
Visited: A



BFS Traversal

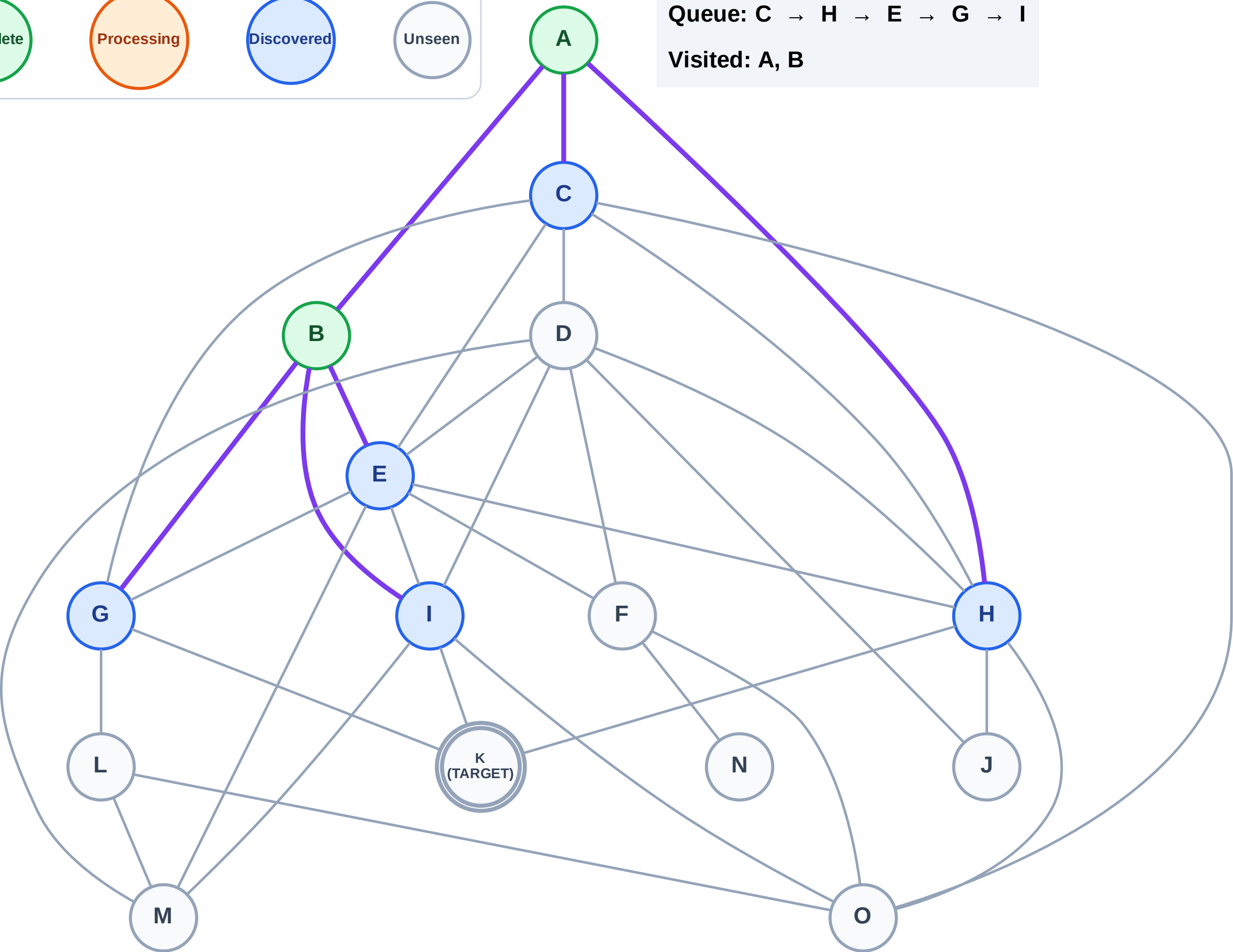
Step 5: Discover E, G, I from B

Legend



Queue: C → H → E → G → I

Visited: A, B



BFS Traversal

Step 6: Process node C

Legend

Complete

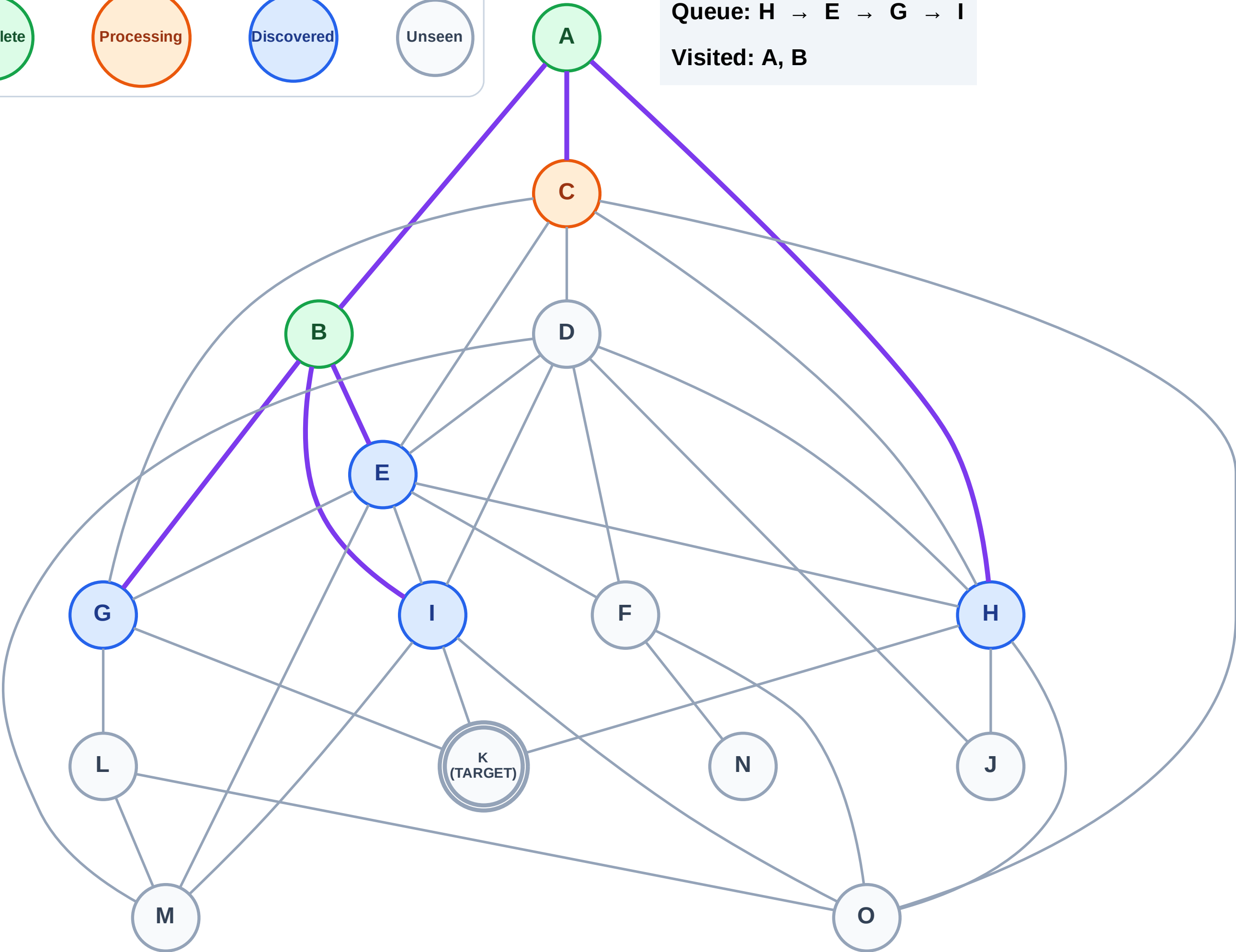
Processing

Discovered

Unseen

Queue: H → E → G → I

Visited: A, B



BFS Traversal

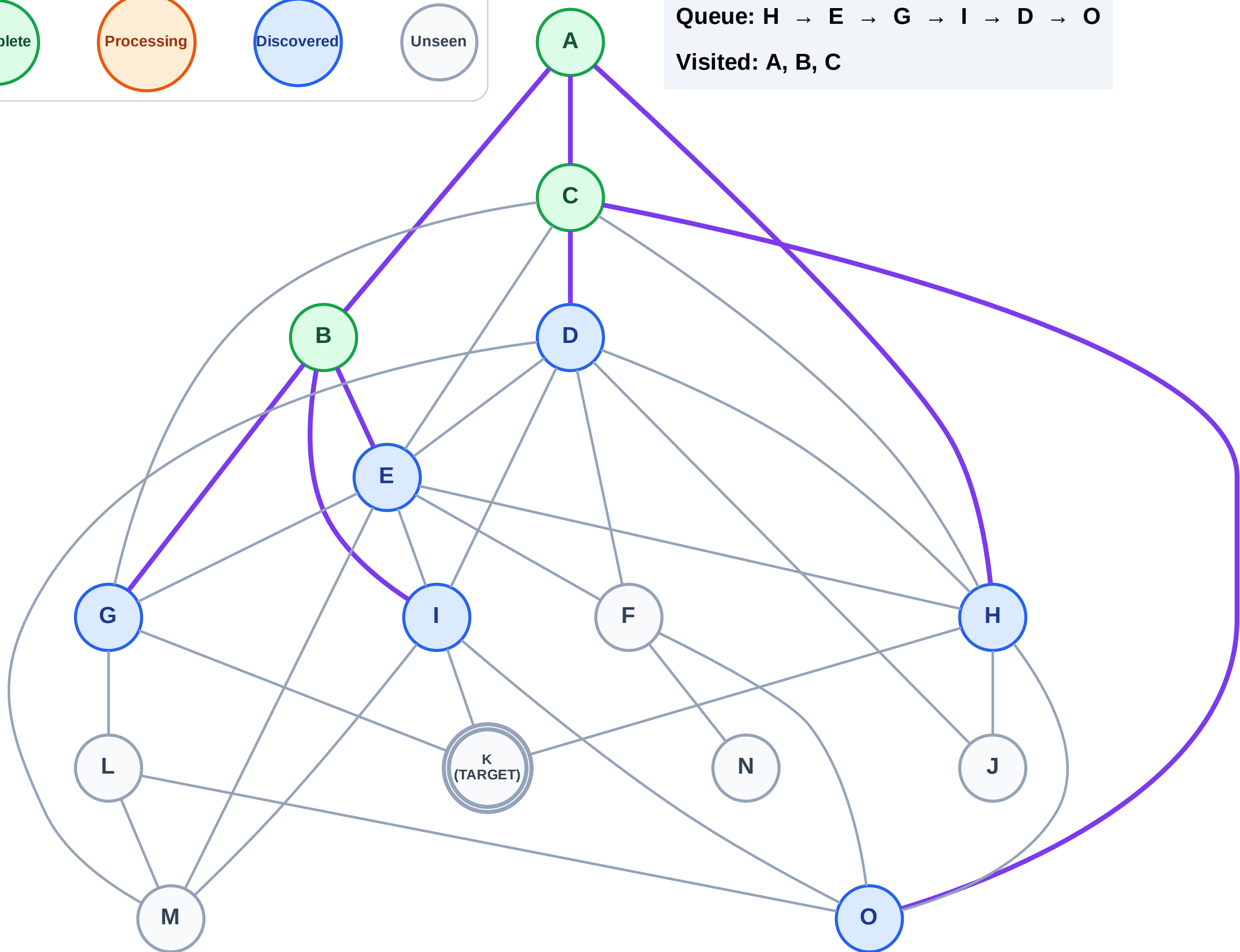
Step 7: Discover D, O from C

Legend



Queue: H → E → G → I → D → O

Visited: A, B, C



BFS Traversal

Step 8: Process node H

Legend

Complete

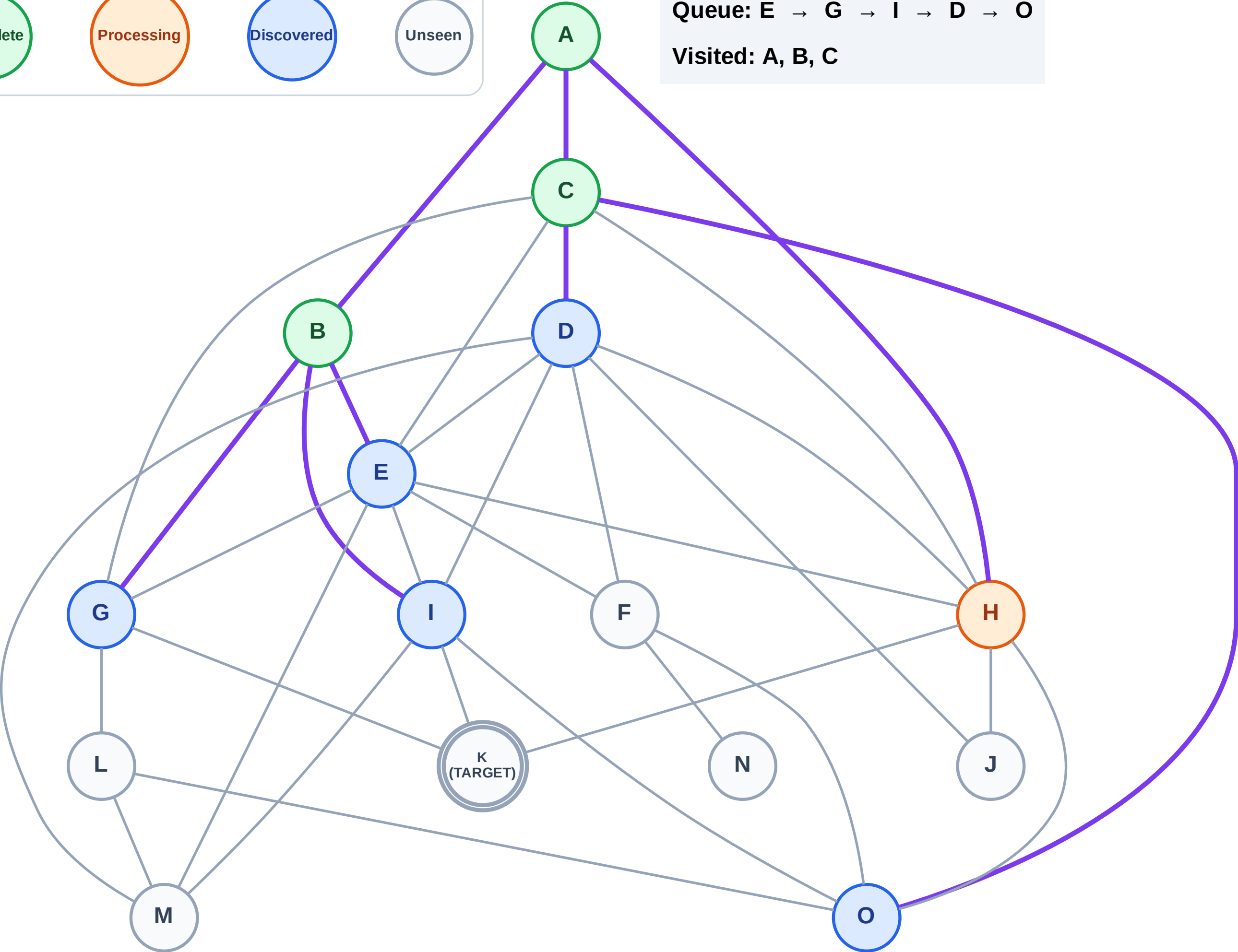
Processing

Discovered

Unseen

Queue: E → G → I → D → O

Visited: A, B, C



BFS Traversal

Step 9: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (2 hops): A → H → K

Legend

Complete

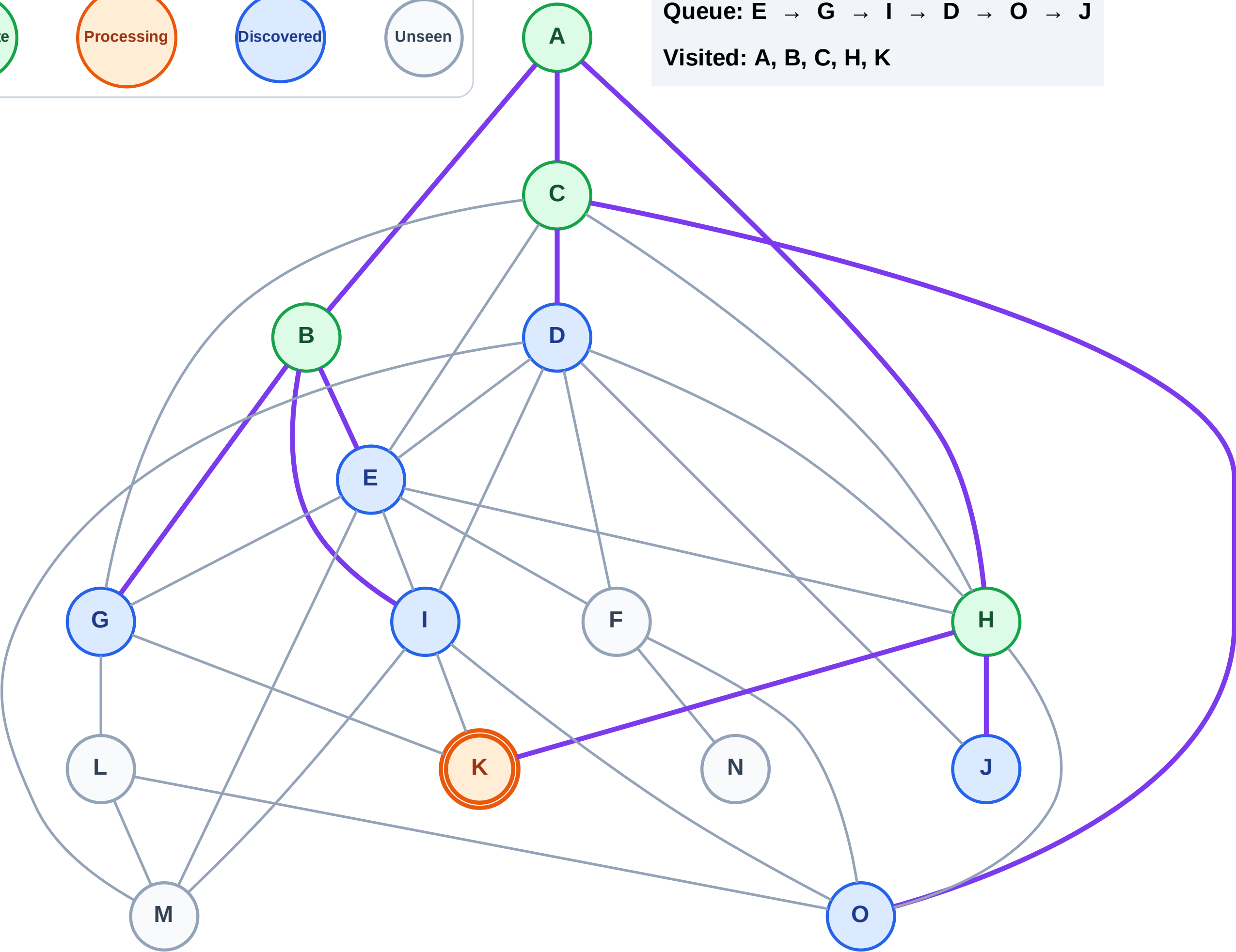
Processing

Discovered

Unseen

Queue: E → G → I → D → O → J

Visited: A, B, C, H, K



DFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend

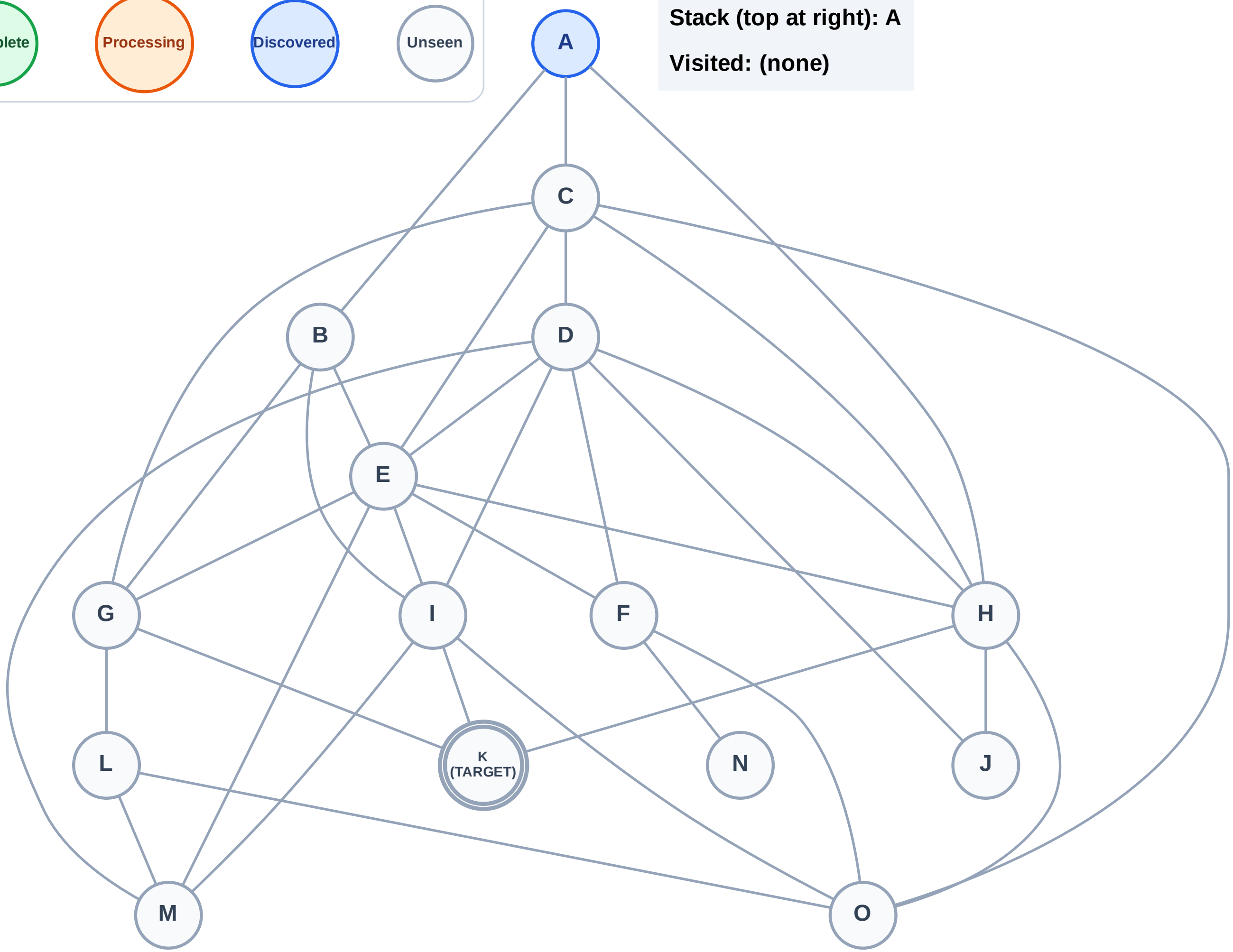
Complete

Processing

Discovered

Unseen

Stack (top at right): A
Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

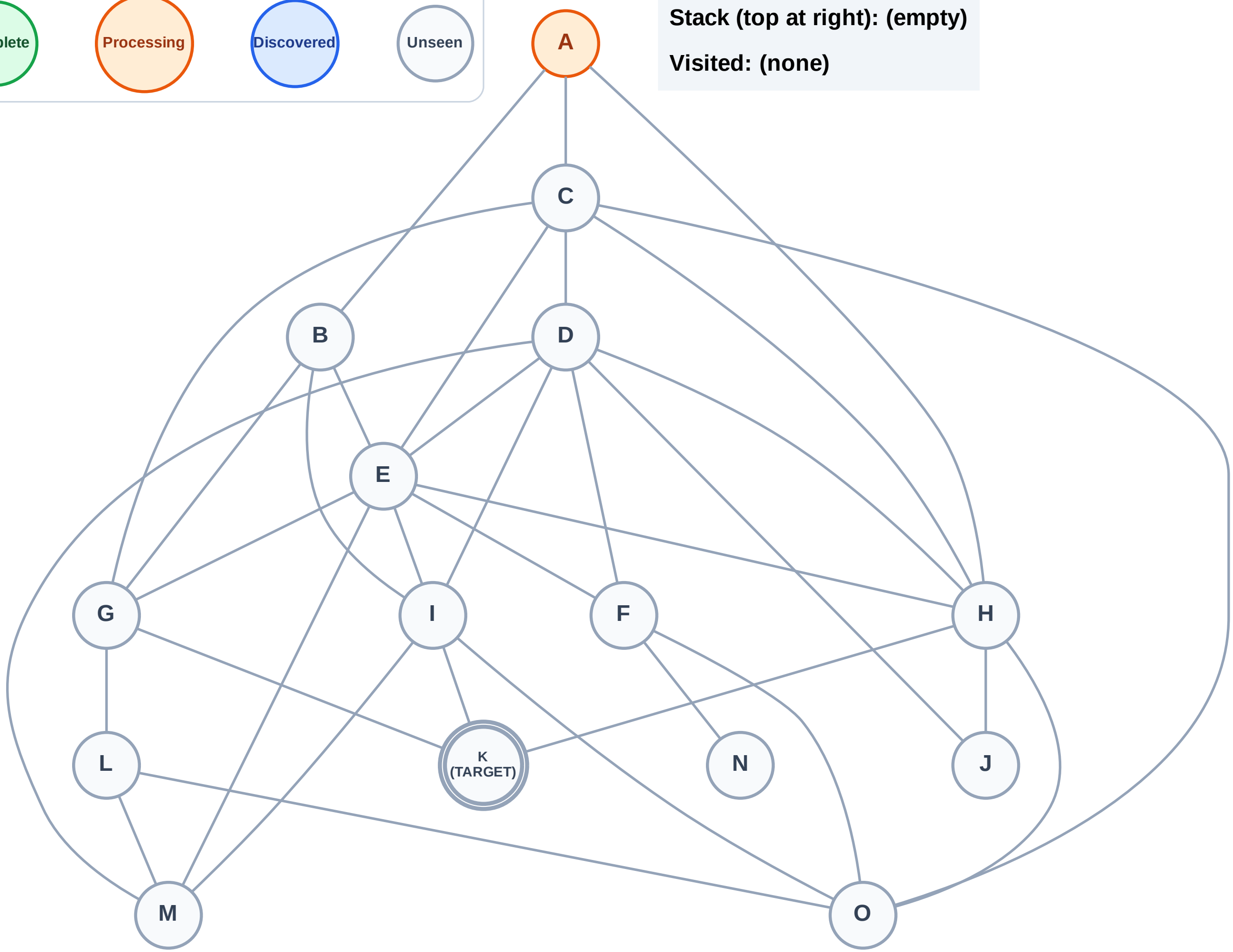
Complete

Processing

Discovered

Unseen

Stack (top at right): (empty)
Visited: (none)



DFS Traversal

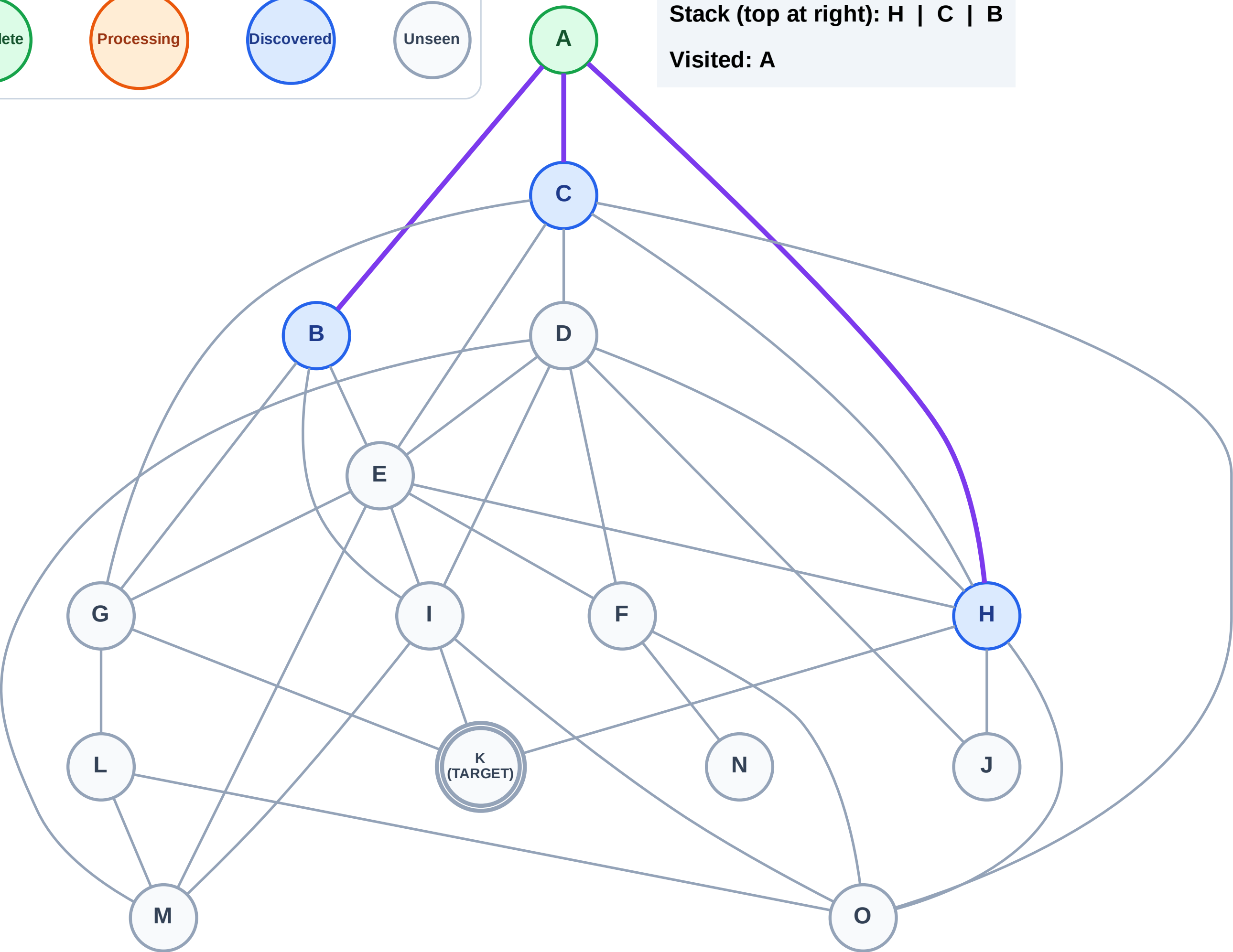
Step 3: Discover H, C, B from A

Legend



Stack (top at right): H | C | B

Visited: A



DFS Traversal

Step 4: Process node B

Legend

Complete

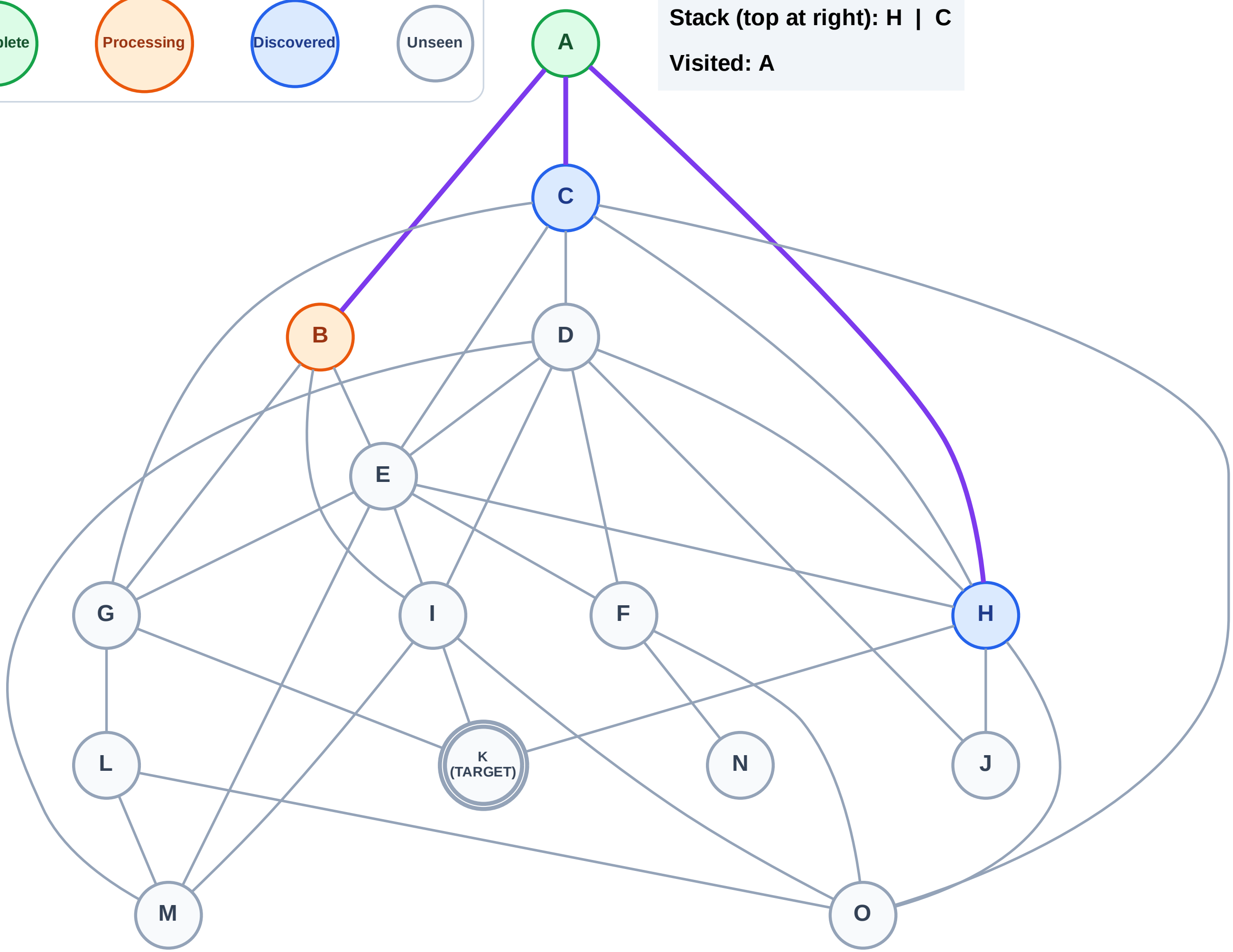
Processing

Discovered

Unseen

Stack (top at right): H | C

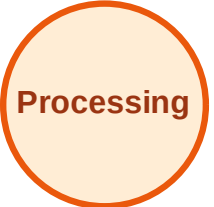
Visited: A



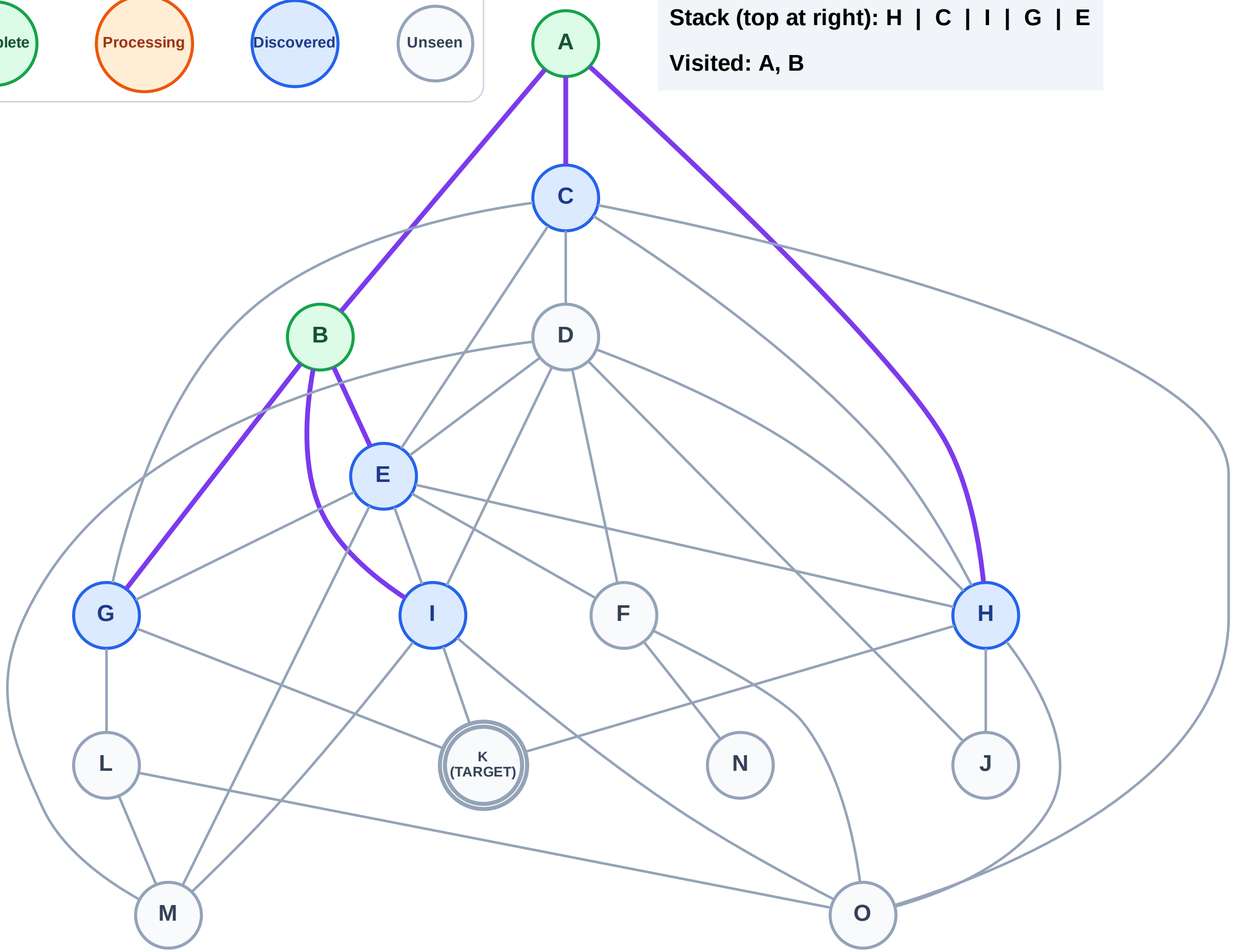
DFS Traversal

Step 5: Discover I, G, E from B

Legend



Stack (top at right): H | C | I | G | E
Visited: A, B



DFS Traversal

Step 6: Process node E

Legend

Complete

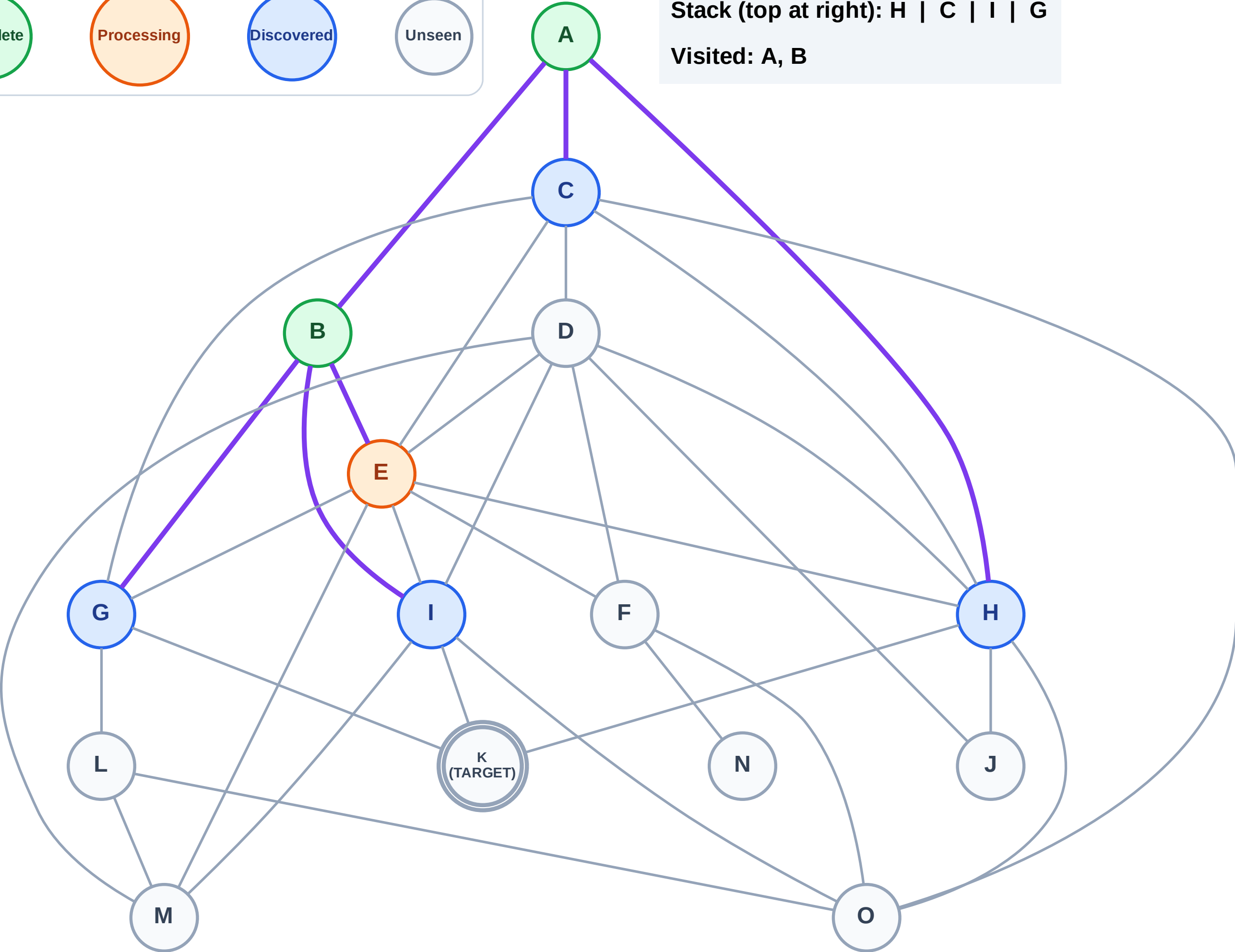
Processing

Discovered

Unseen

Stack (top at right): H | C | I | G

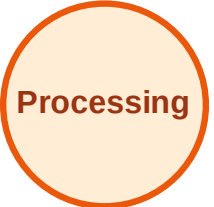
Visited: A, B



DFS Traversal

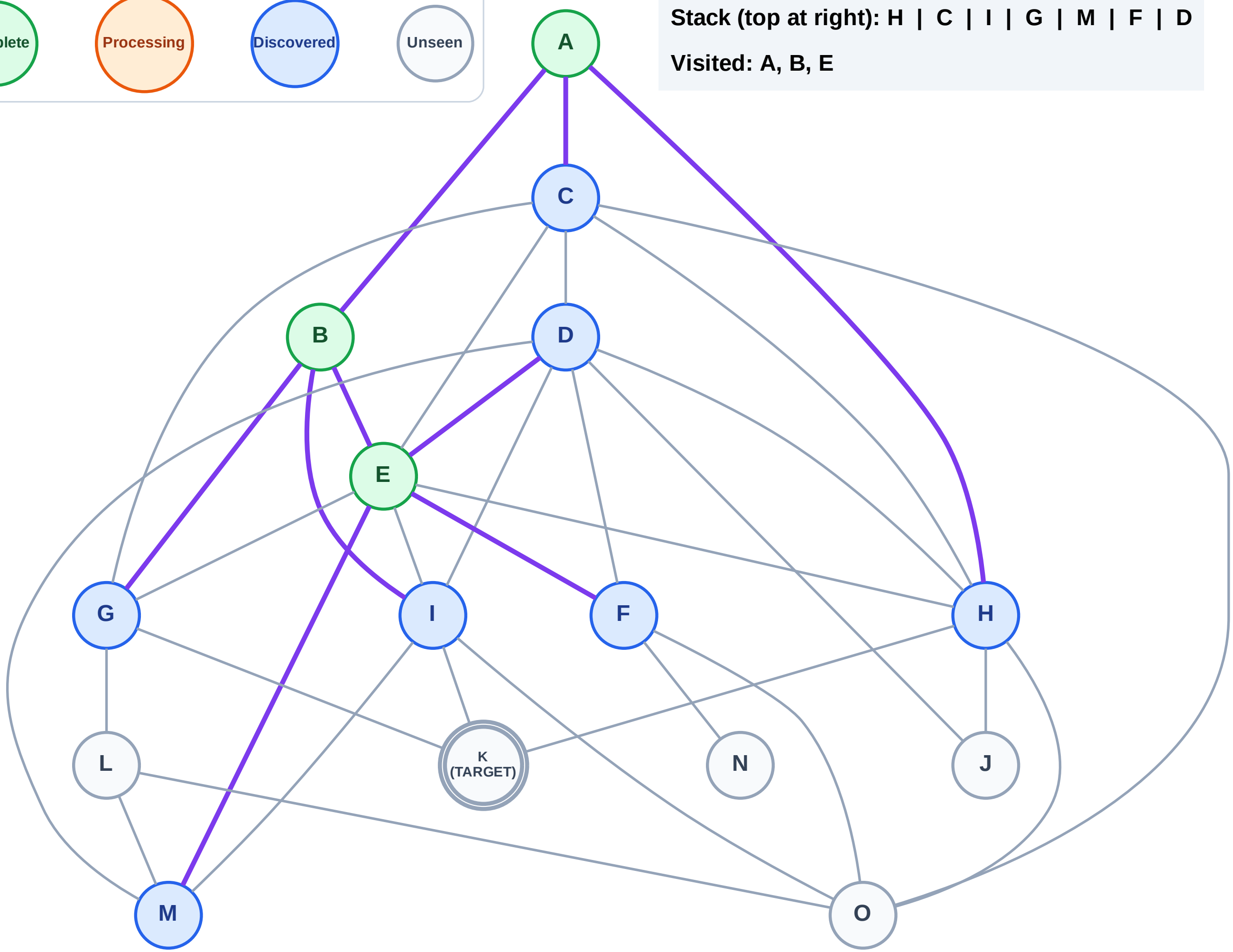
Step 7: Discover M, F, D from E

Legend



Stack (top at right): H | C | I | G | M | F | D

Visited: A, B, E



DFS Traversal

Step 8: Process node D

Legend

Complete

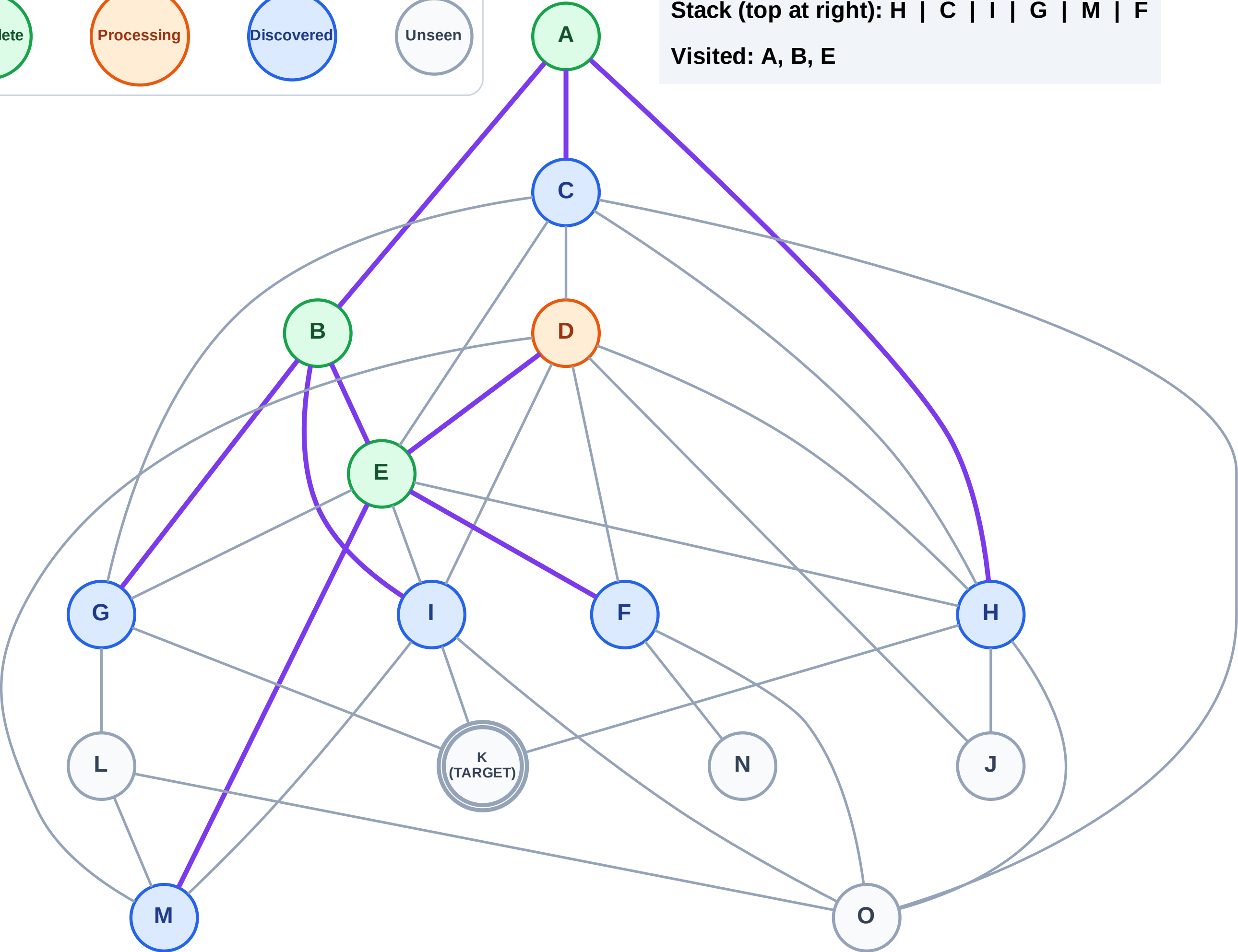
Processing

Discovered

Unseen

Stack (top at right): H | C | I | G | M | F

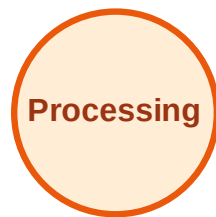
Visited: A, B, E



DFS Traversal

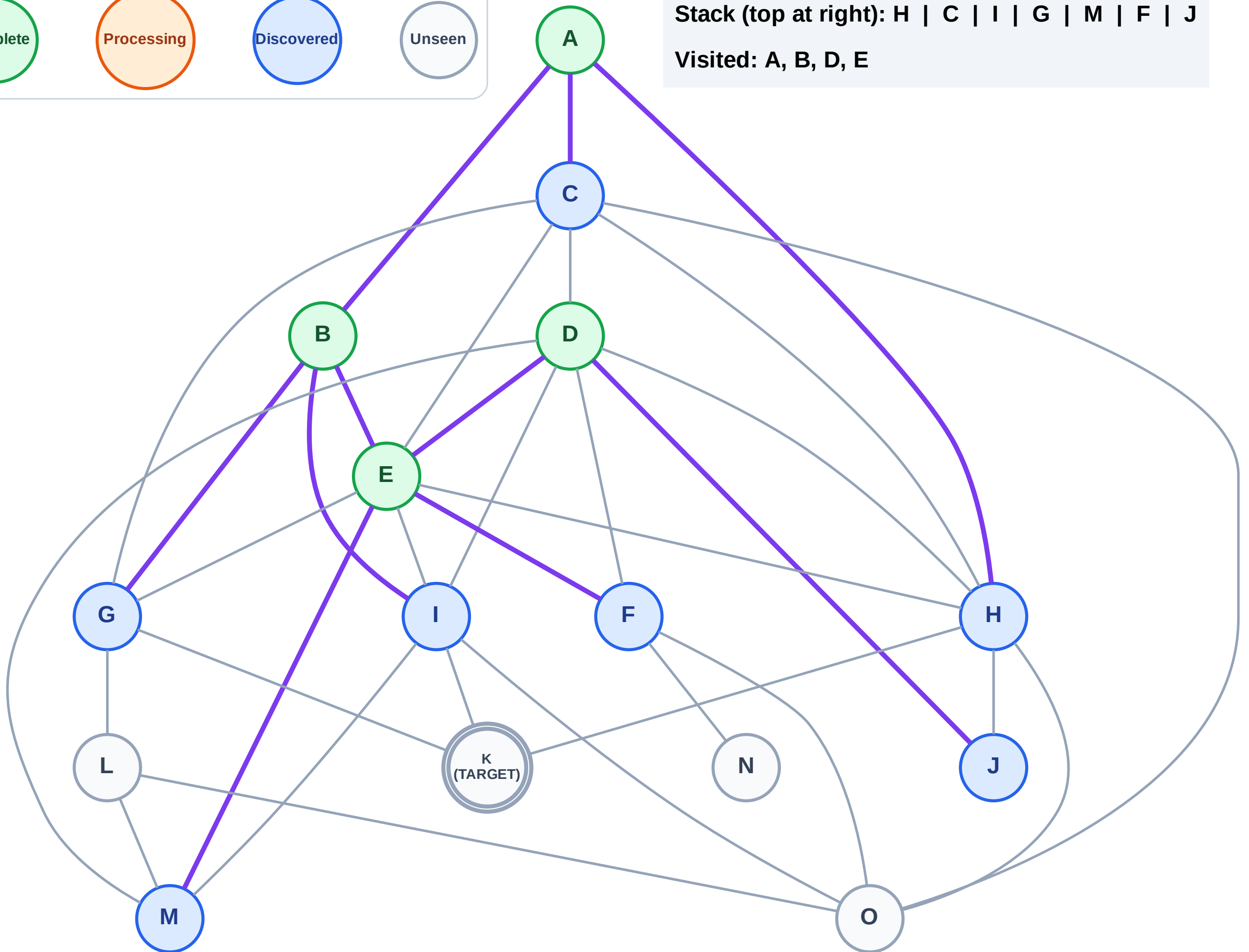
Step 9: Discover J from D

Legend



Stack (top at right): H | C | I | G | M | F | J

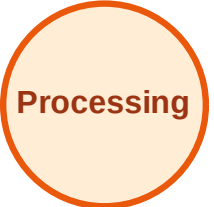
Visited: A, B, D, E



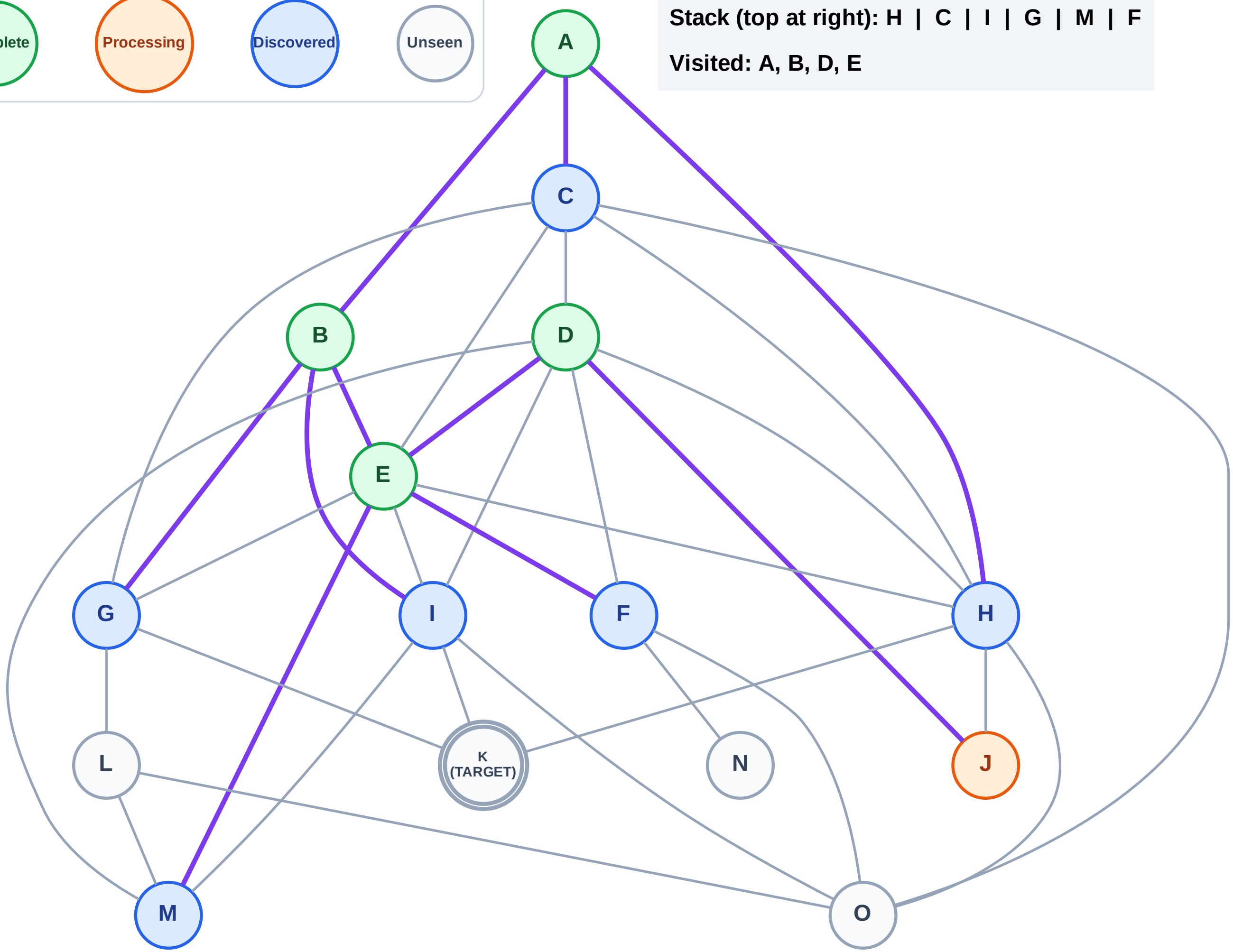
DFS Traversal

Step 10: Process node J

Legend



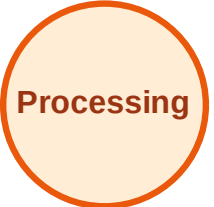
Stack (top at right): H | C | I | G | M | F
Visited: A, B, D, E



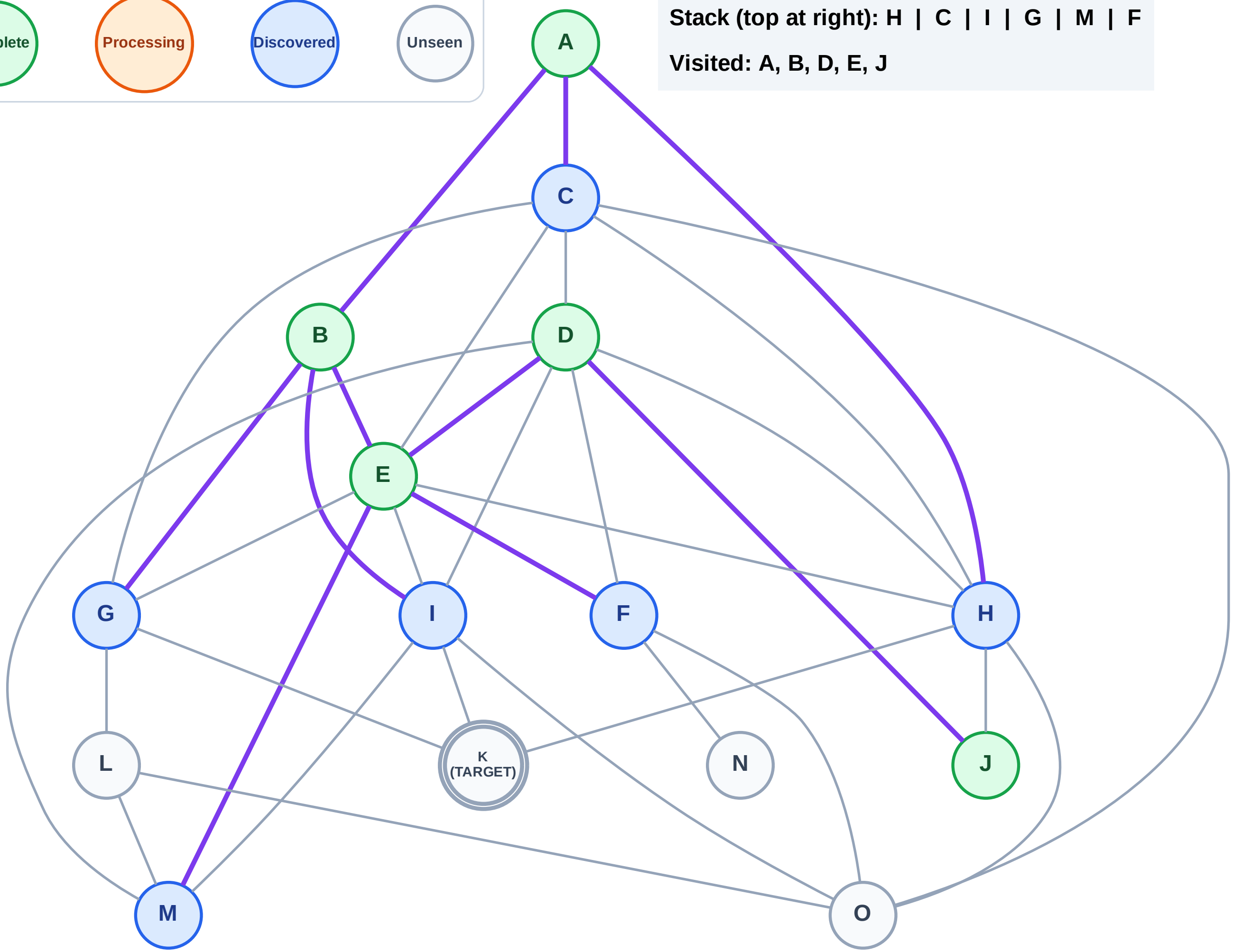
DFS Traversal

Step 11: No new neighbors from J

Legend



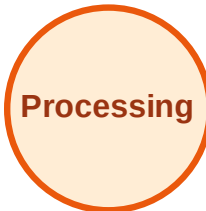
Stack (top at right): H | C | I | G | M | F
Visited: A, B, D, E, J



DFS Traversal

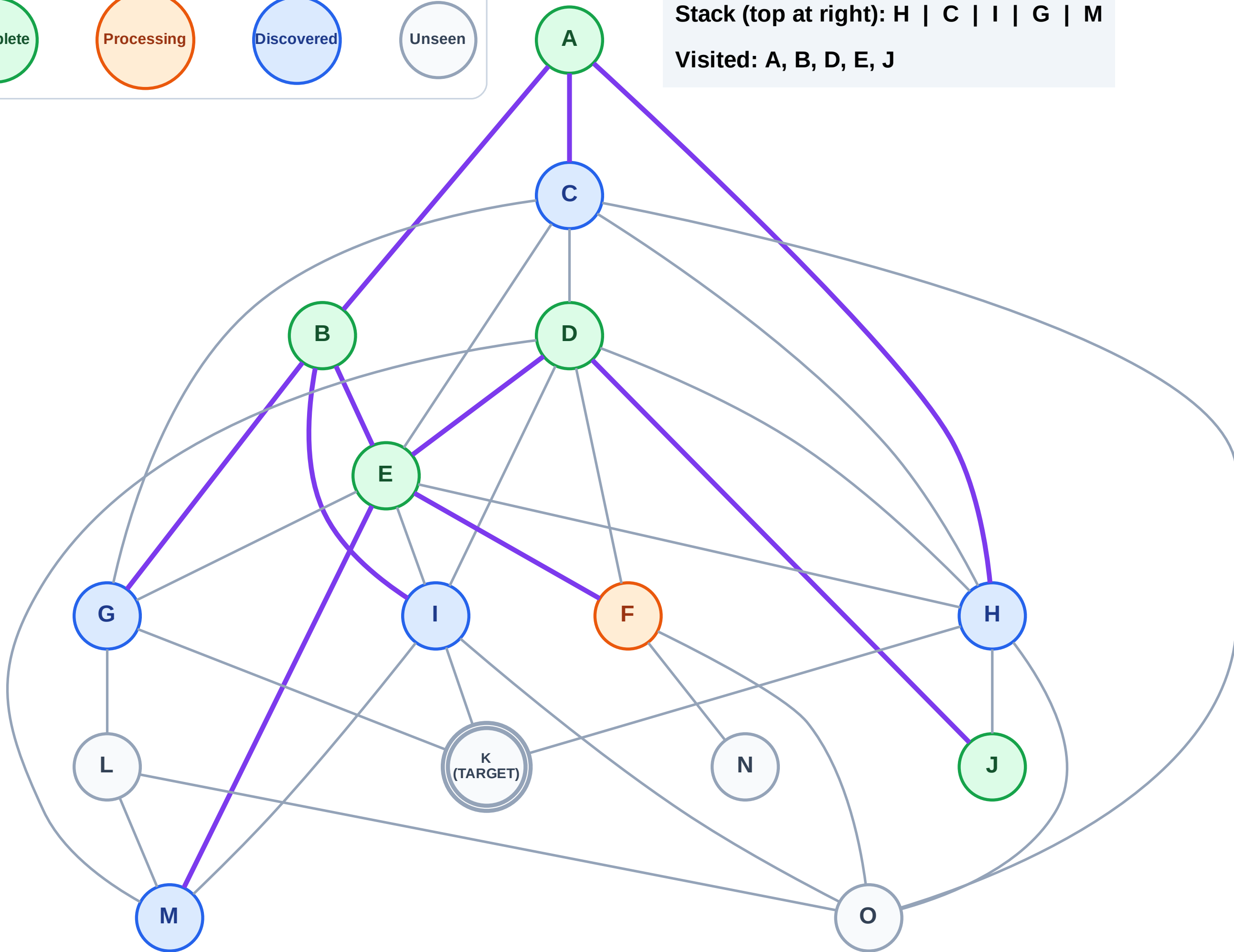
Step 12: Process node F

Legend



Stack (top at right): H | C | I | G | M

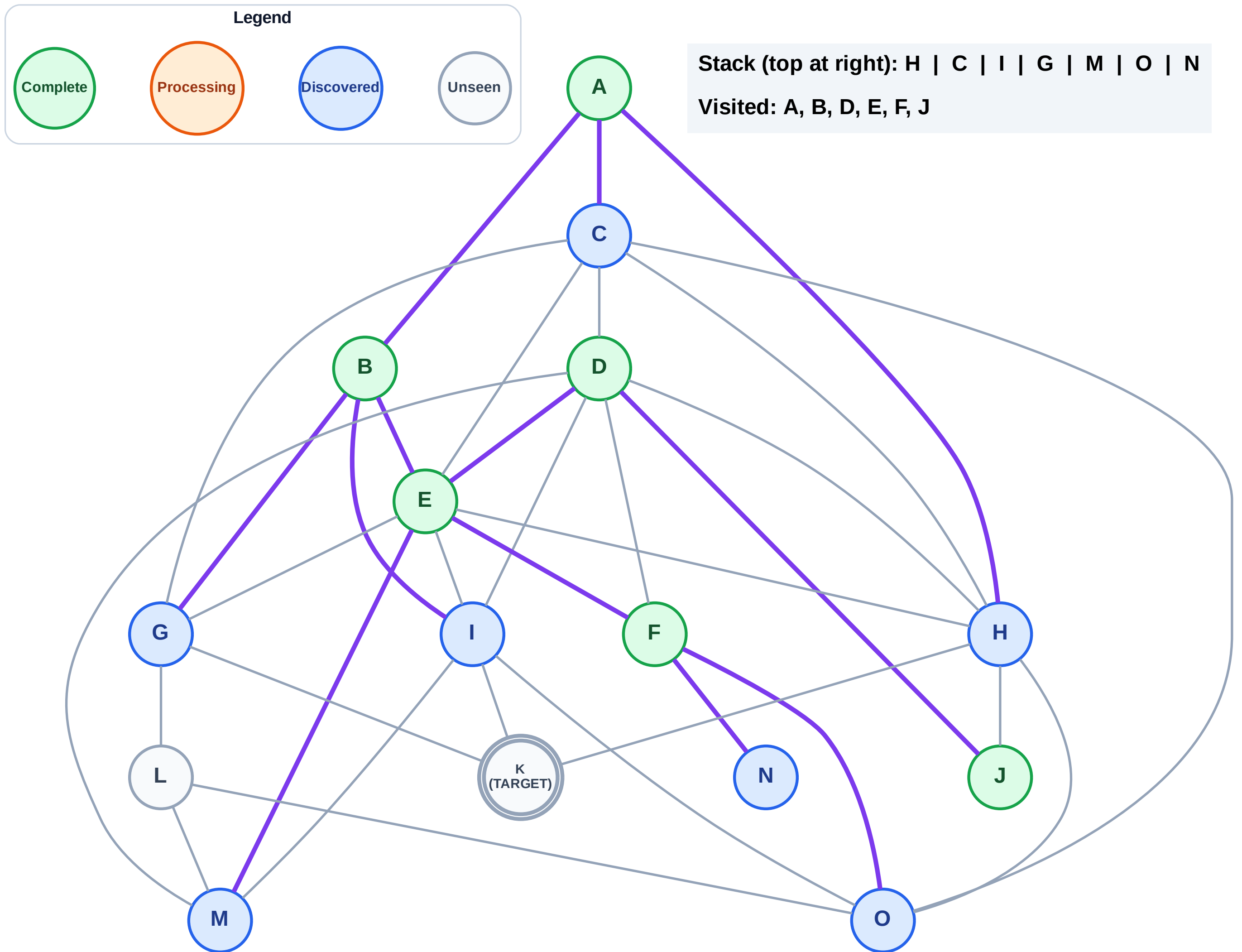
Visited: A, B, D, E, J



Step 13: Discover O, N from F

Stack (top at right): H | C | I | G | M | O | N

Visited: A, B, D, E, F, J



DFS Traversal

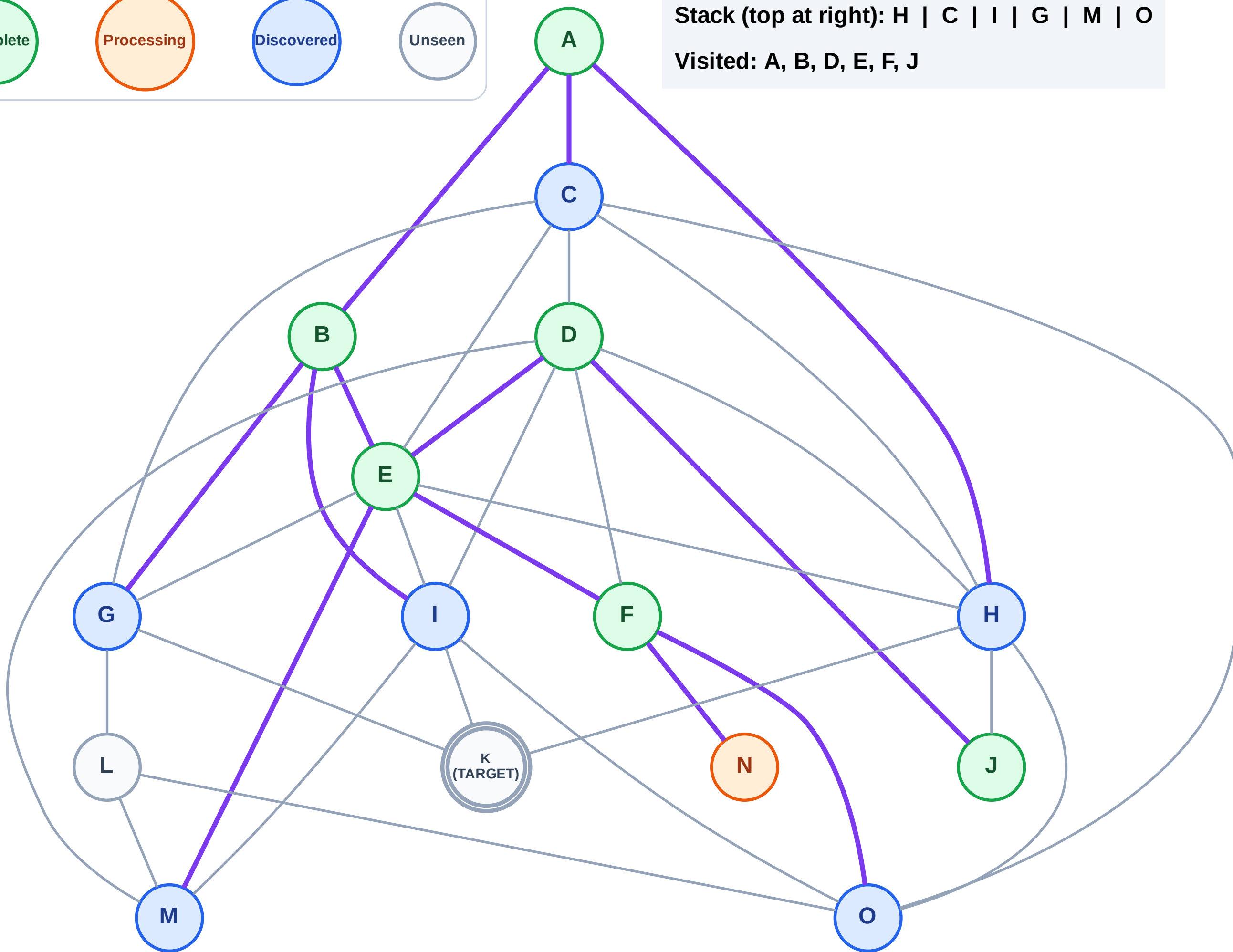
Step 14: Process node N

Legend



Stack (top at right): H | C | I | G | M | O

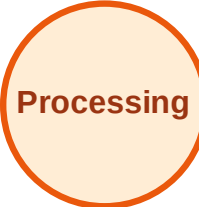
Visited: A, B, D, E, F, J



DFS Traversal

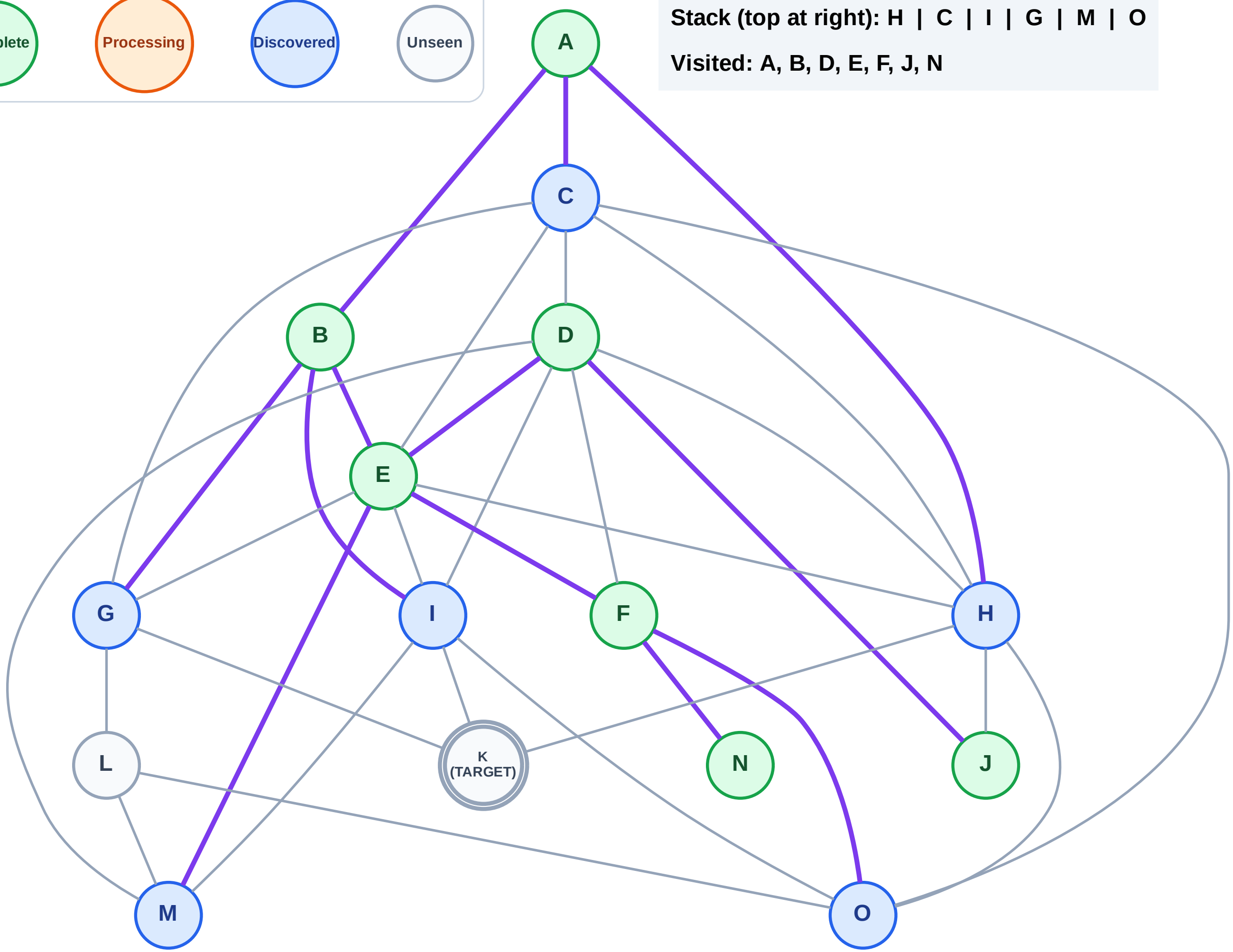
Step 15: No new neighbors from N

Legend



Stack (top at right): H | C | I | G | M | O

Visited: A, B, D, E, F, J, N



DFS Traversal

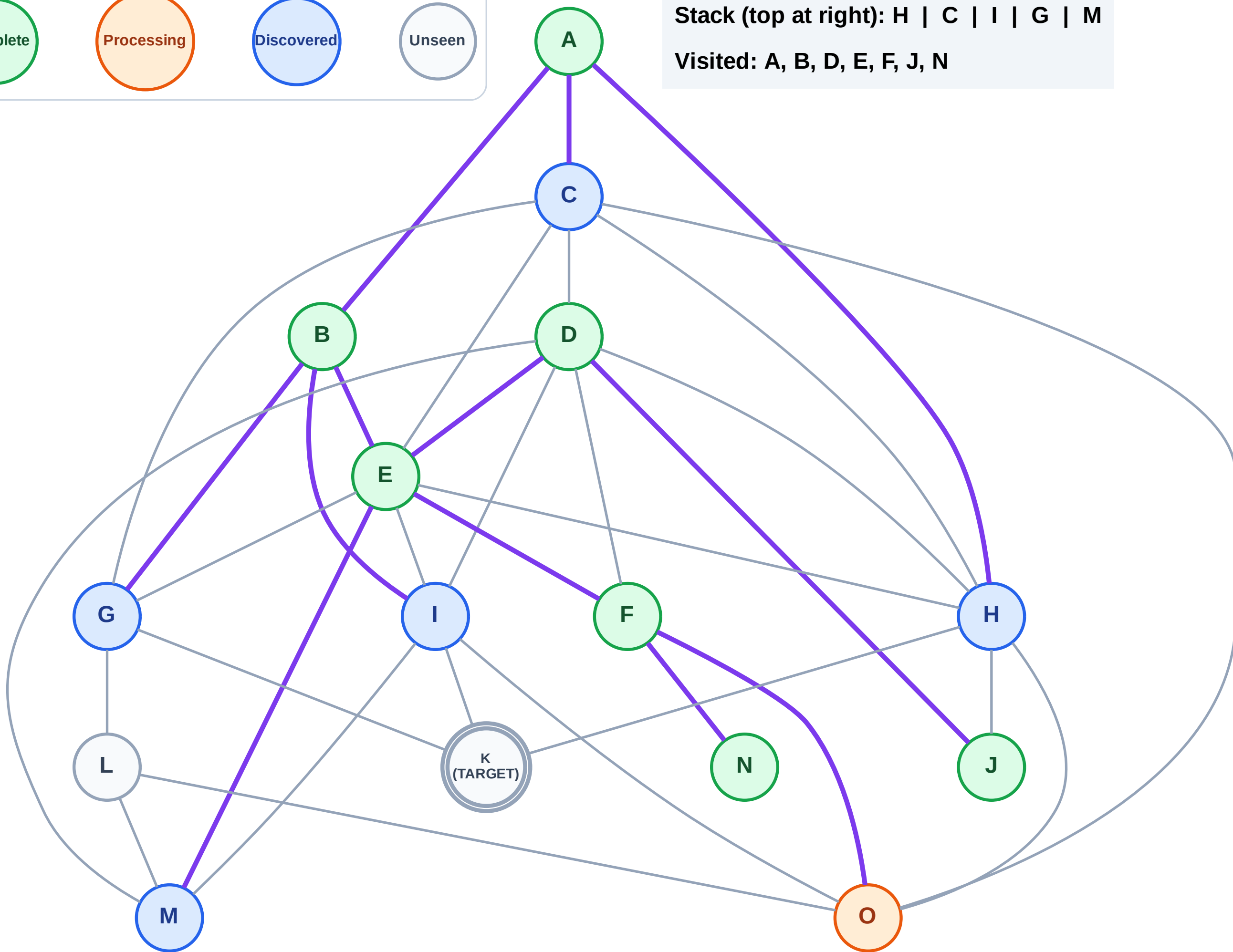
Step 16: Process node O

Legend



Stack (top at right): H | C | I | G | M

Visited: A, B, D, E, F, J, N



DFS Traversal

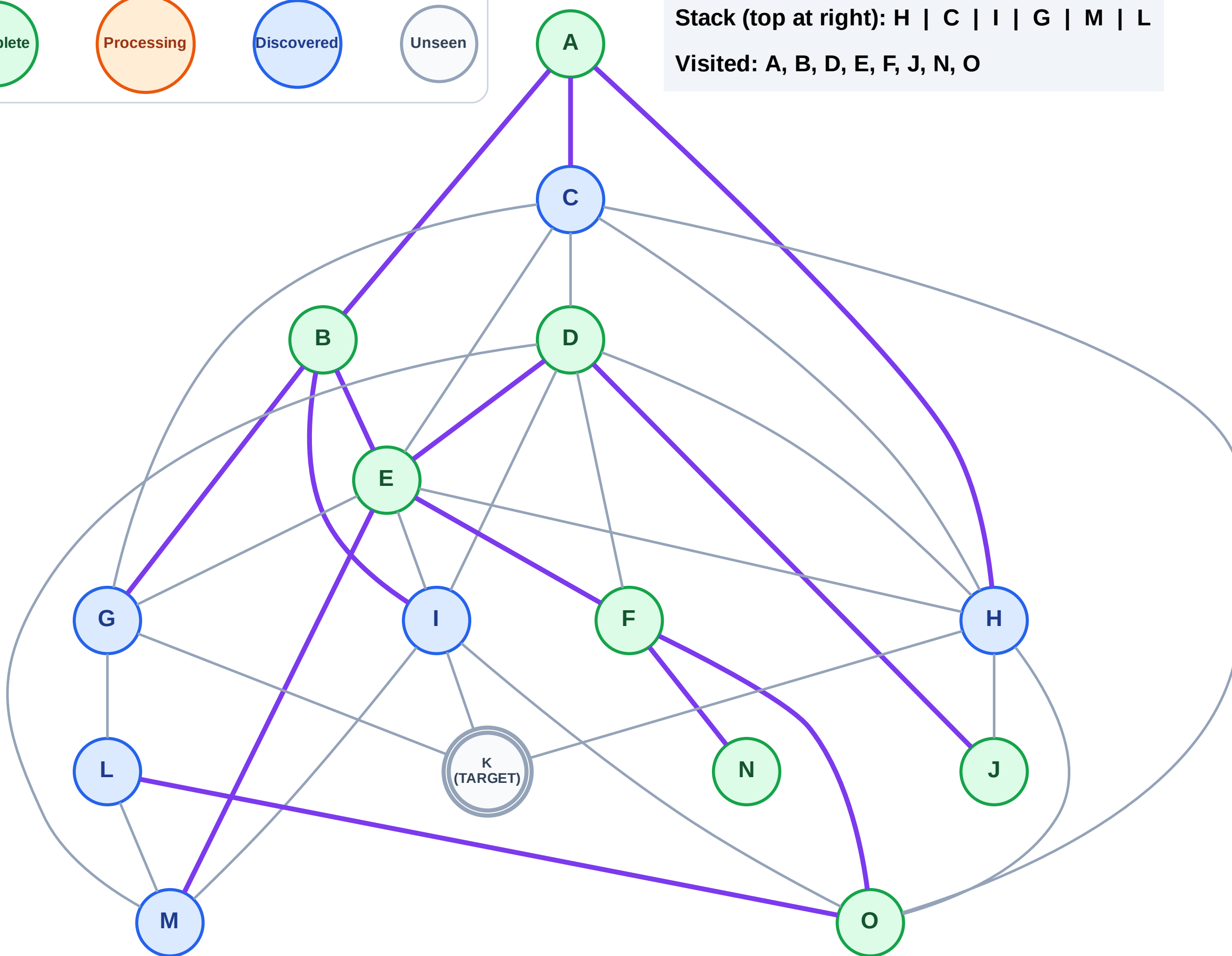
Step 17: Discover L from O

Legend



Stack (top at right): H | C | I | G | M | L

Visited: A, B, D, E, F, J, N, O



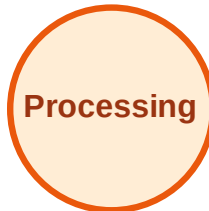
DFS Traversal

Step 18: Process node L

Legend



Complete



Processing



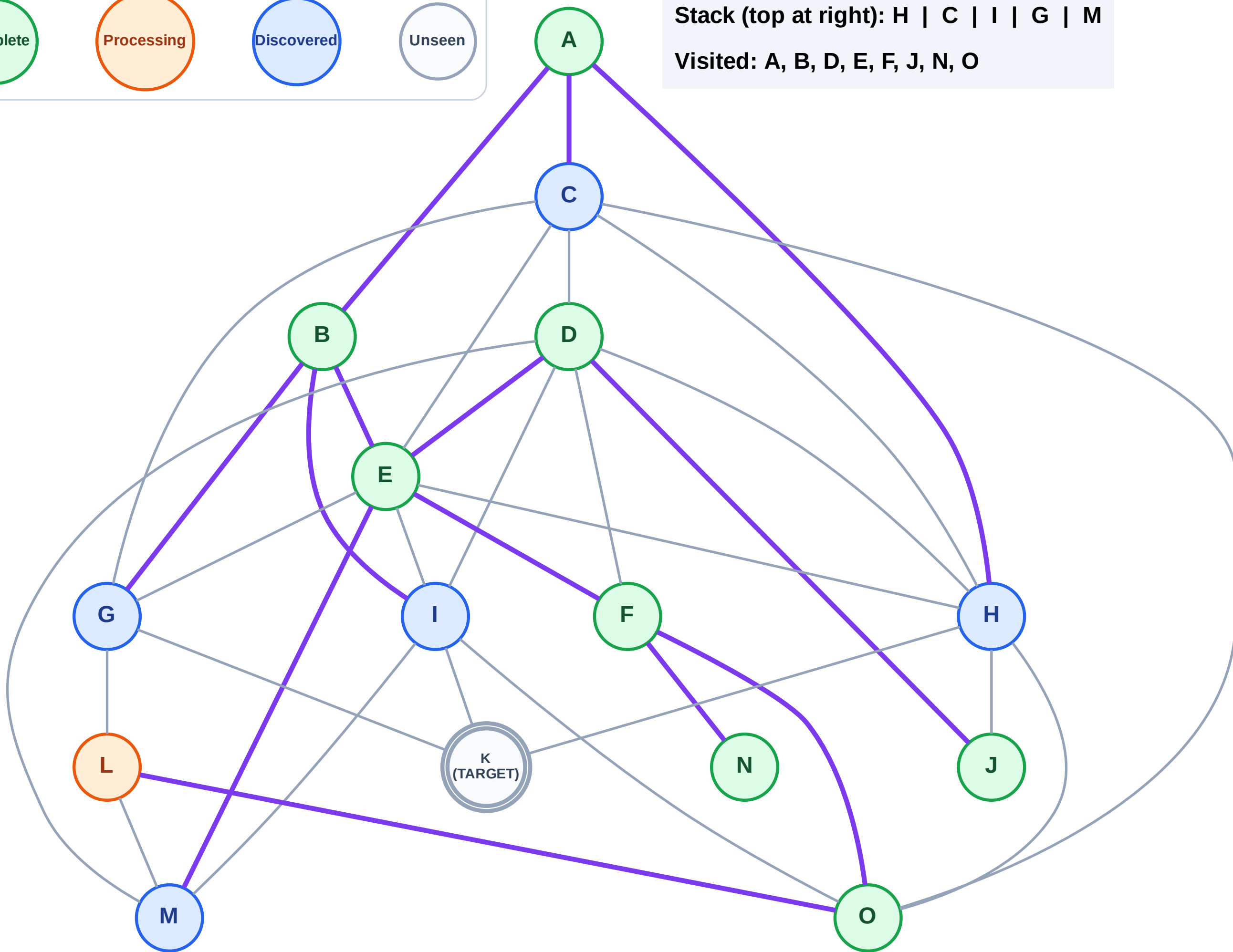
Discovered



Unseen

Stack (top at right): H | C | I | G | M

Visited: A, B, D, E, F, J, N, O



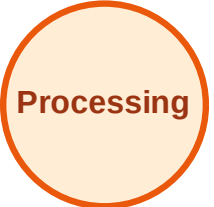
DFS Traversal

Step 19: No new neighbors from L

Legend



Complete



Processing



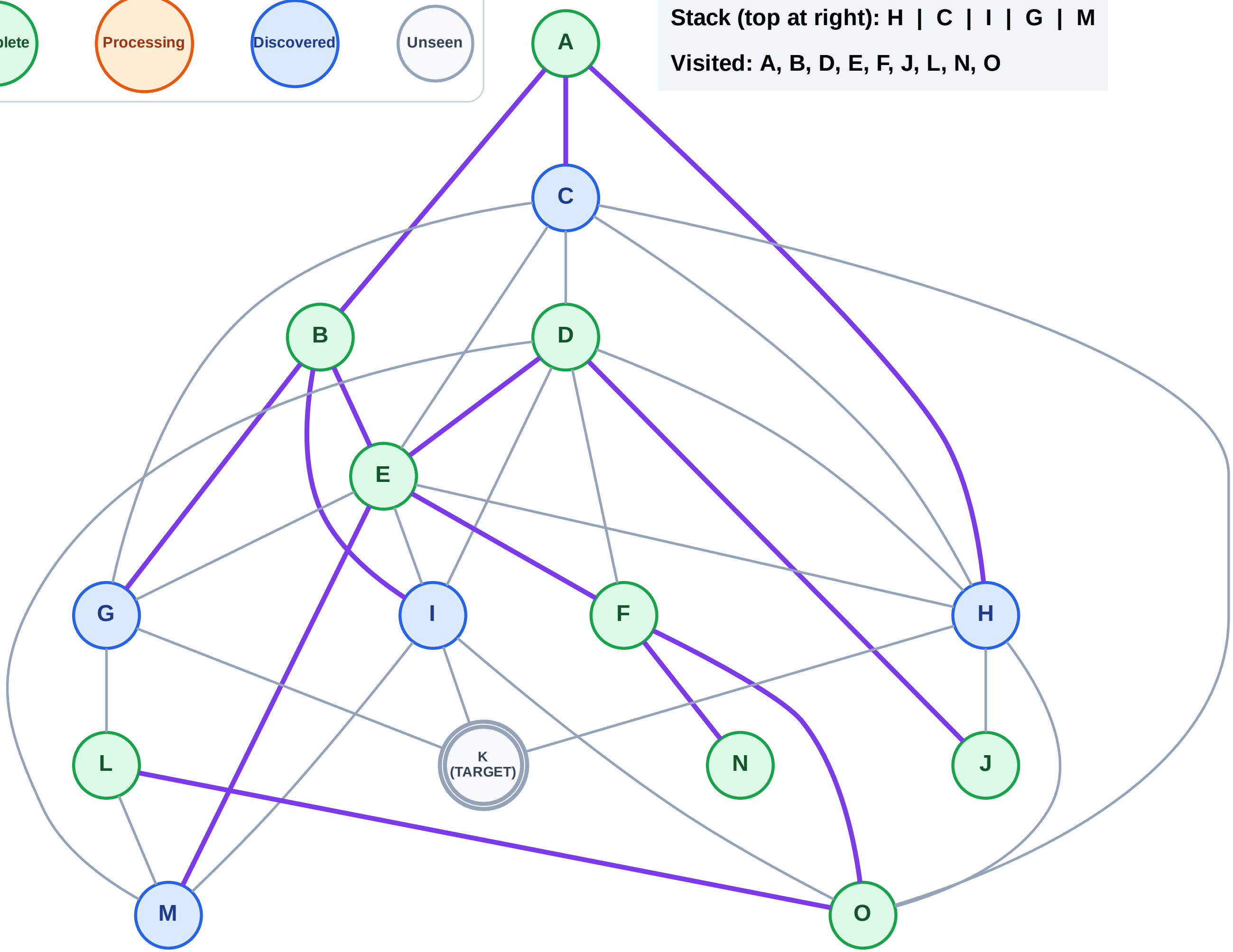
Discovered



Unseen

Stack (top at right): H | C | I | G | M

Visited: A, B, D, E, F, J, L, N, O



DFS Traversal

Step 20: Process node M

Legend

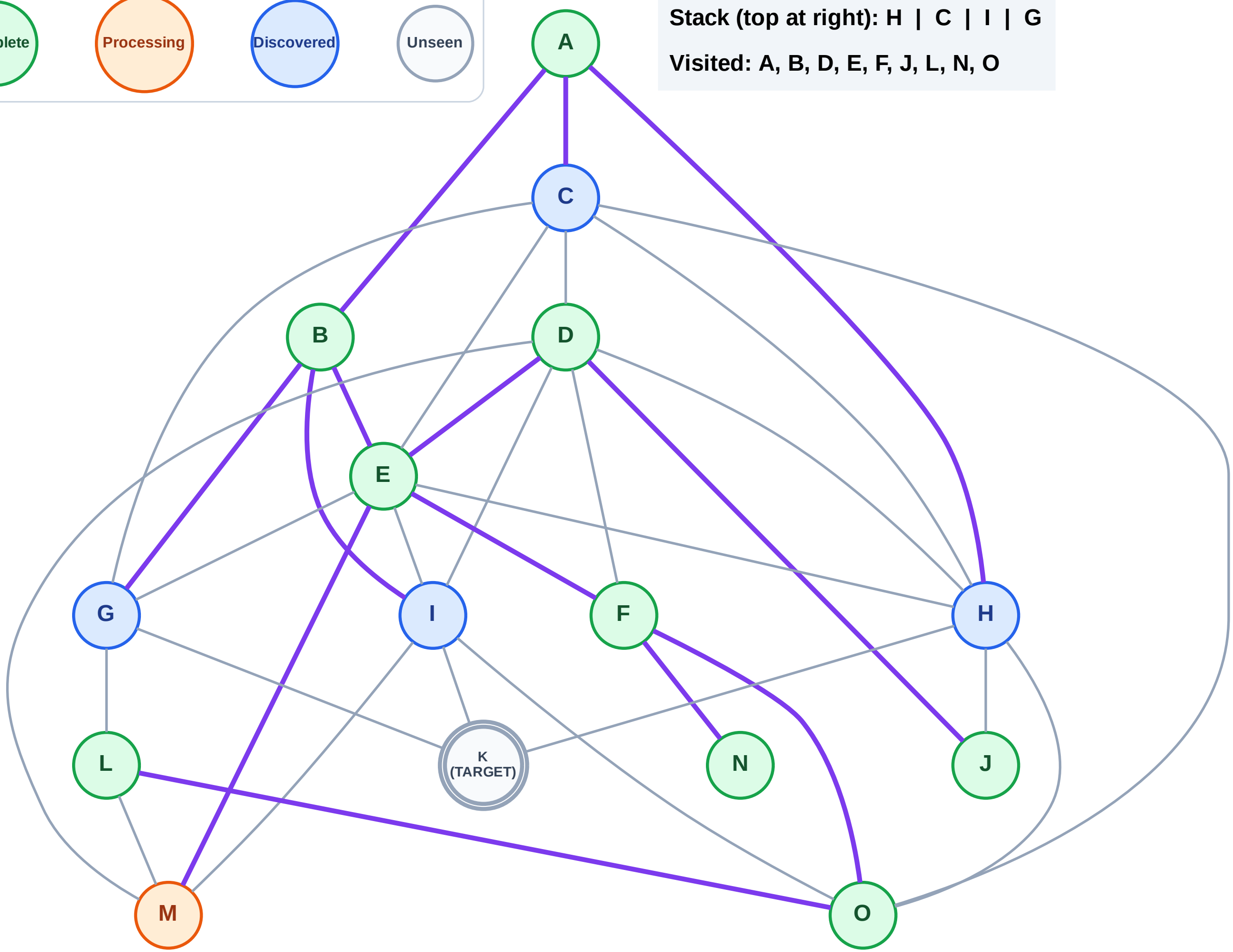
Complete

Processing

Discovered

Unseen

Stack (top at right): H | C | I | G
Visited: A, B, D, E, F, J, L, N, O



DFS Traversal

Step 21: No new neighbors from M

Legend

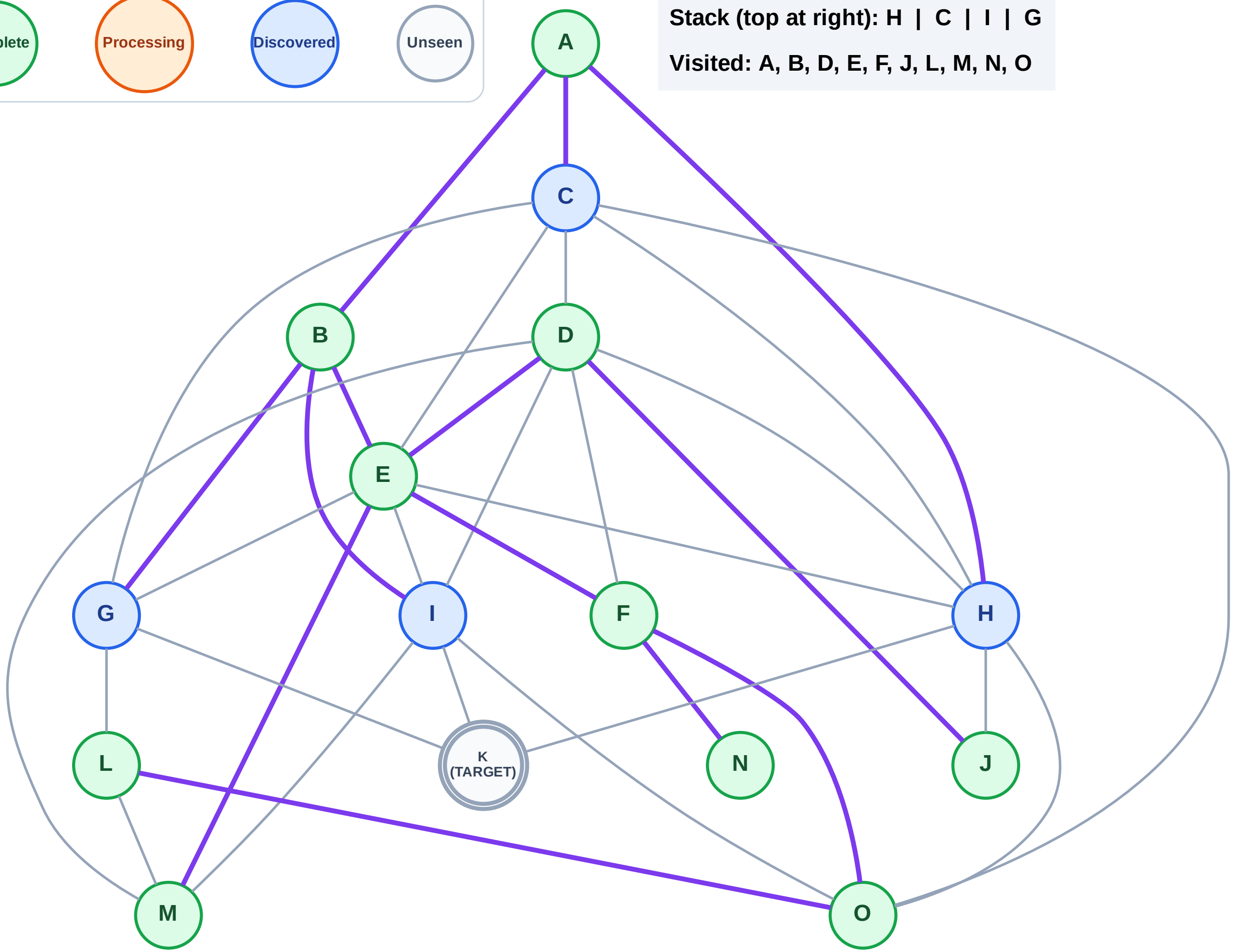
Complete

Processing

Discovered

Unseen

Stack (top at right): H | C | I | G
Visited: A, B, D, E, F, J, L, M, N, O



DFS Traversal

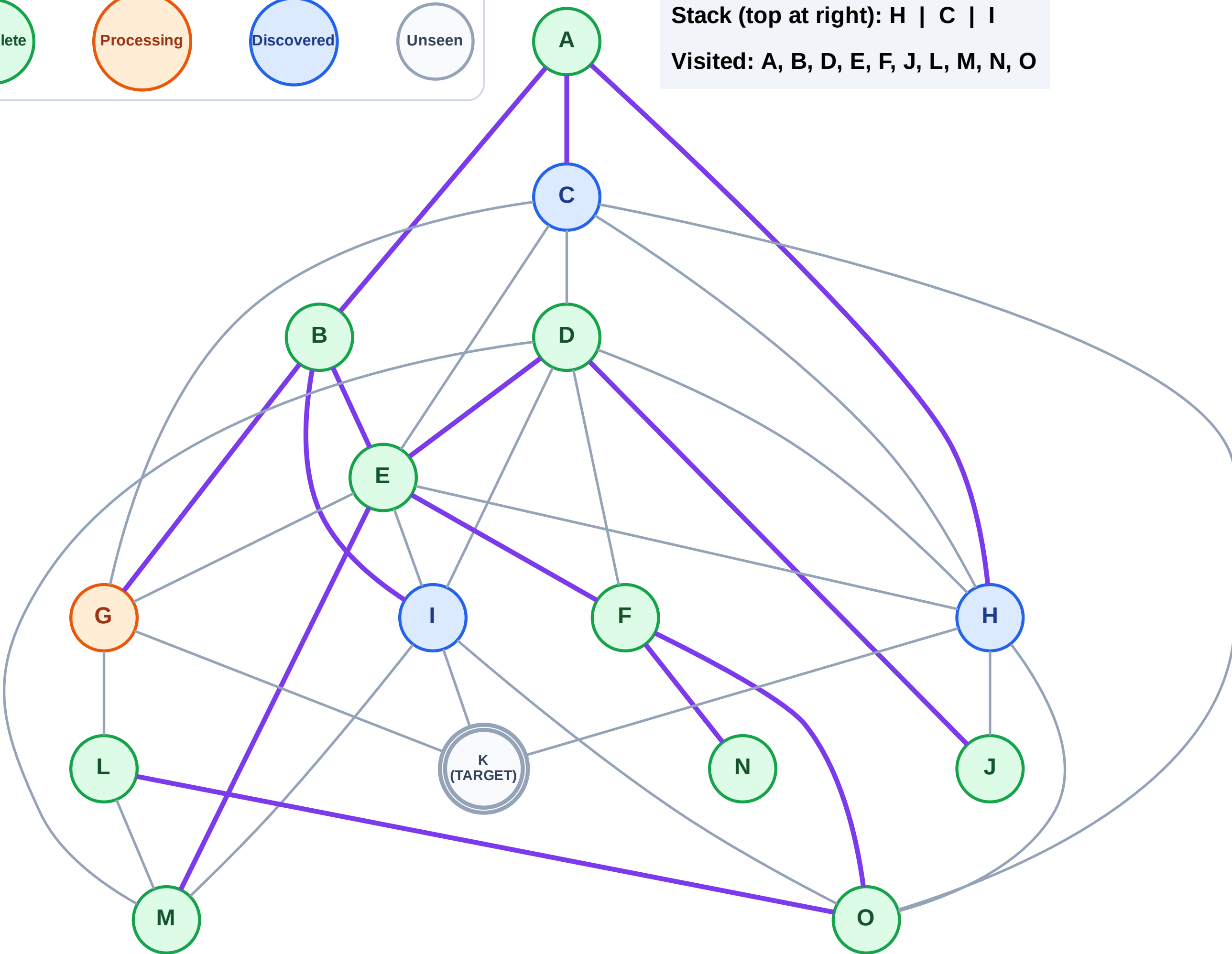
Step 22: Process node G

Legend



Stack (top at right): H | C | I

Visited: A, B, D, E, F, J, L, M, N, O



DFS Traversal

Step 23: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (3 hops): A → B → G → K

Legend



Complete



Processing



Discovered



Unseen

Stack (top at right): H | C | I

Visited: A, B, D, E, F, G, J, K, L, M, N, O

